



NovaBike Under Attack

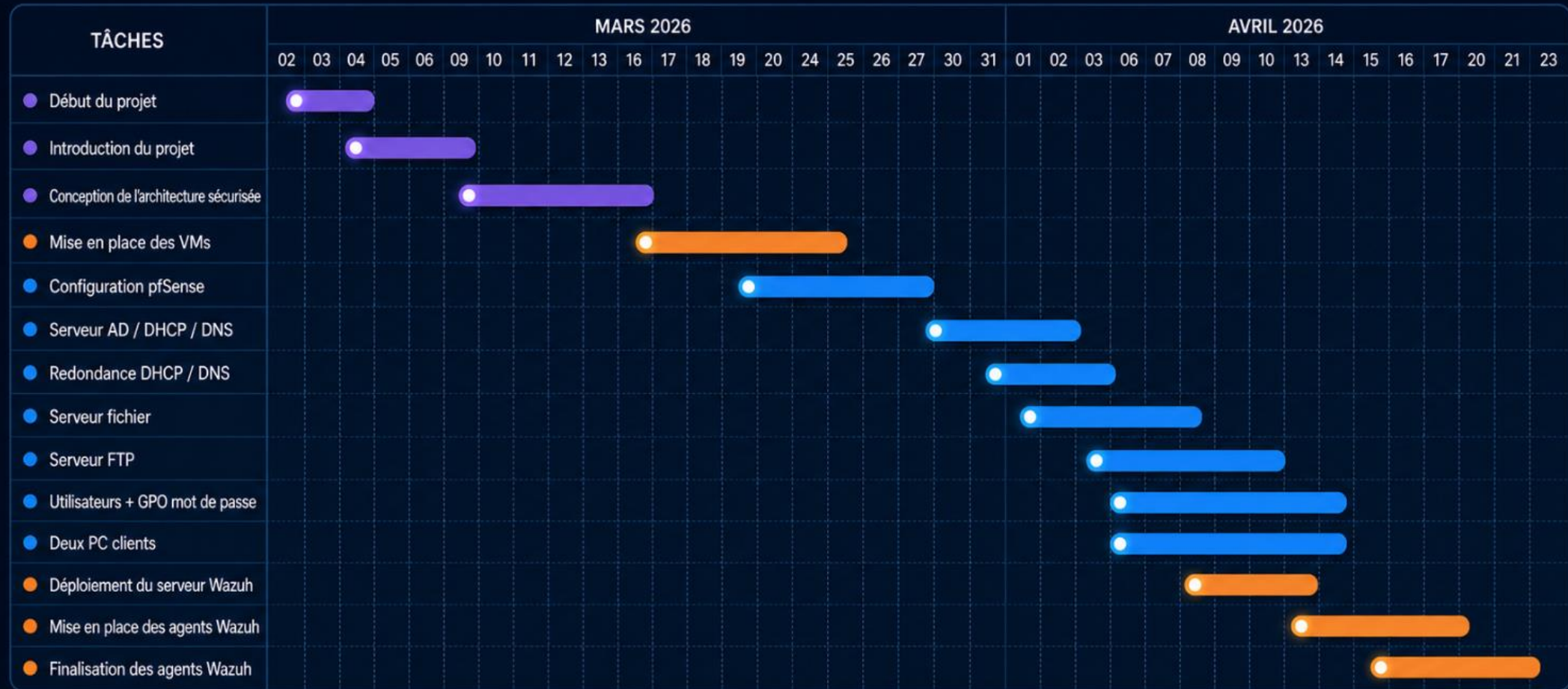
DÉPLOYER, VERROUILLER ET SAUVER UNE INFRASTRUCTURE EN ROUE LIBRE

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Projet INT12 - Diagramme de Gantt



PÉRIODE DU PROJET

03/03/2026 → 23/04/2026



Augustin



Samuel



Tâches communes

Contexte du projet

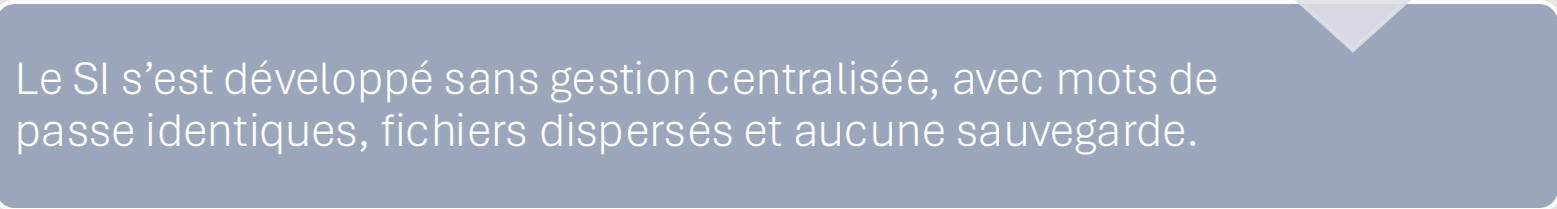
NovaBike est une PME fictive spécialisée dans la vente et la maintenance de vélos électriques.



L'entreprise compte 10 salariés avec atelier, boutique, administration, commercial et gérant.



Le SI s'est développé sans gestion centralisée, avec mots de passe identiques, fichiers dispersés et aucune sauvegarde.





Déployer une infrastructure locale virtualisée avec AD, DNS, DHCP, partage de fichiers et pare-feu.



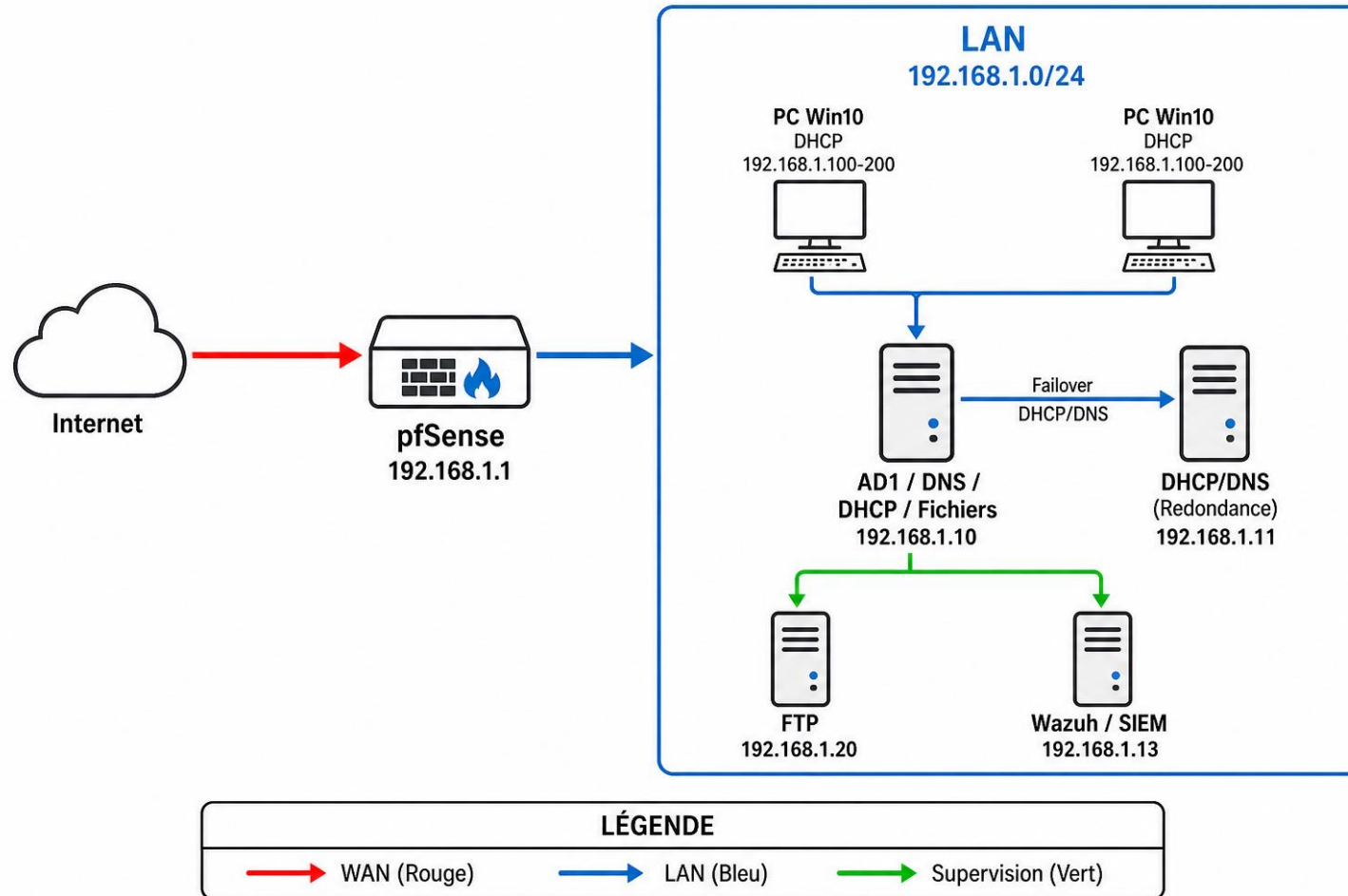
Mettre en place des pratiques de durcissement de base : mots de passe, journalisation, sauvegardes.



Centraliser l'administration et améliorer la résilience de l'infrastructure.

Objectifs du projet

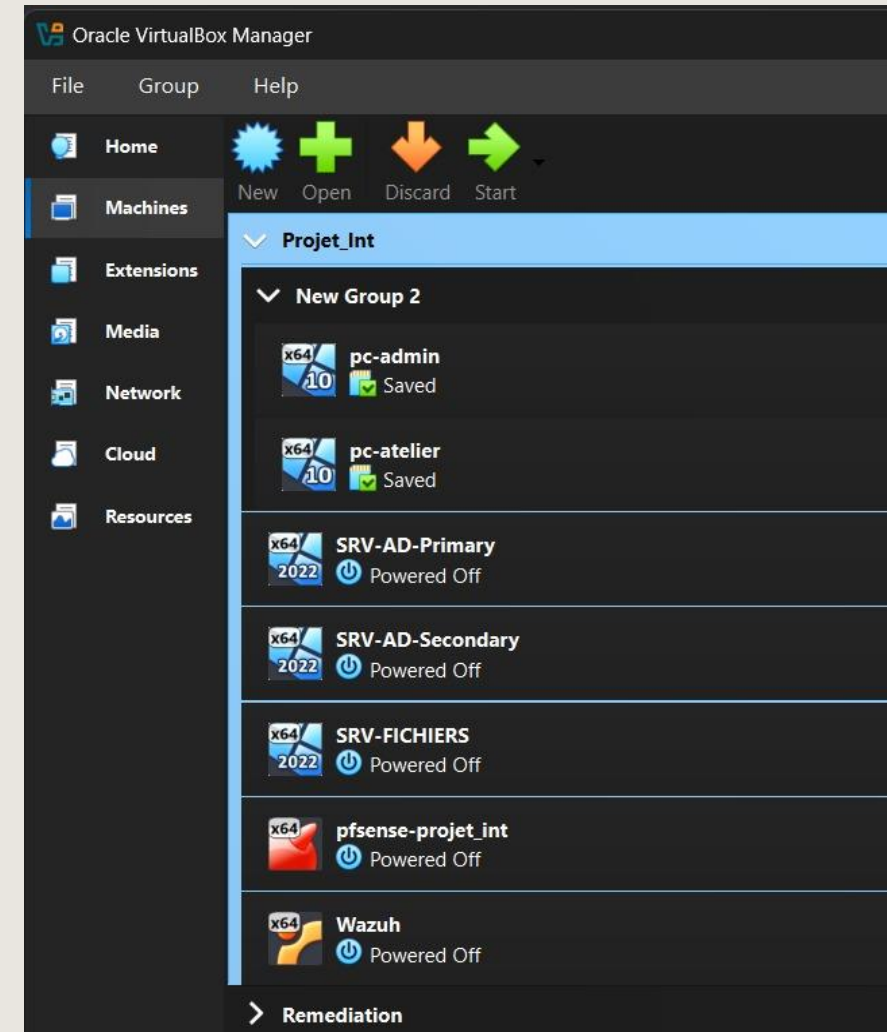
Architecture mise en place



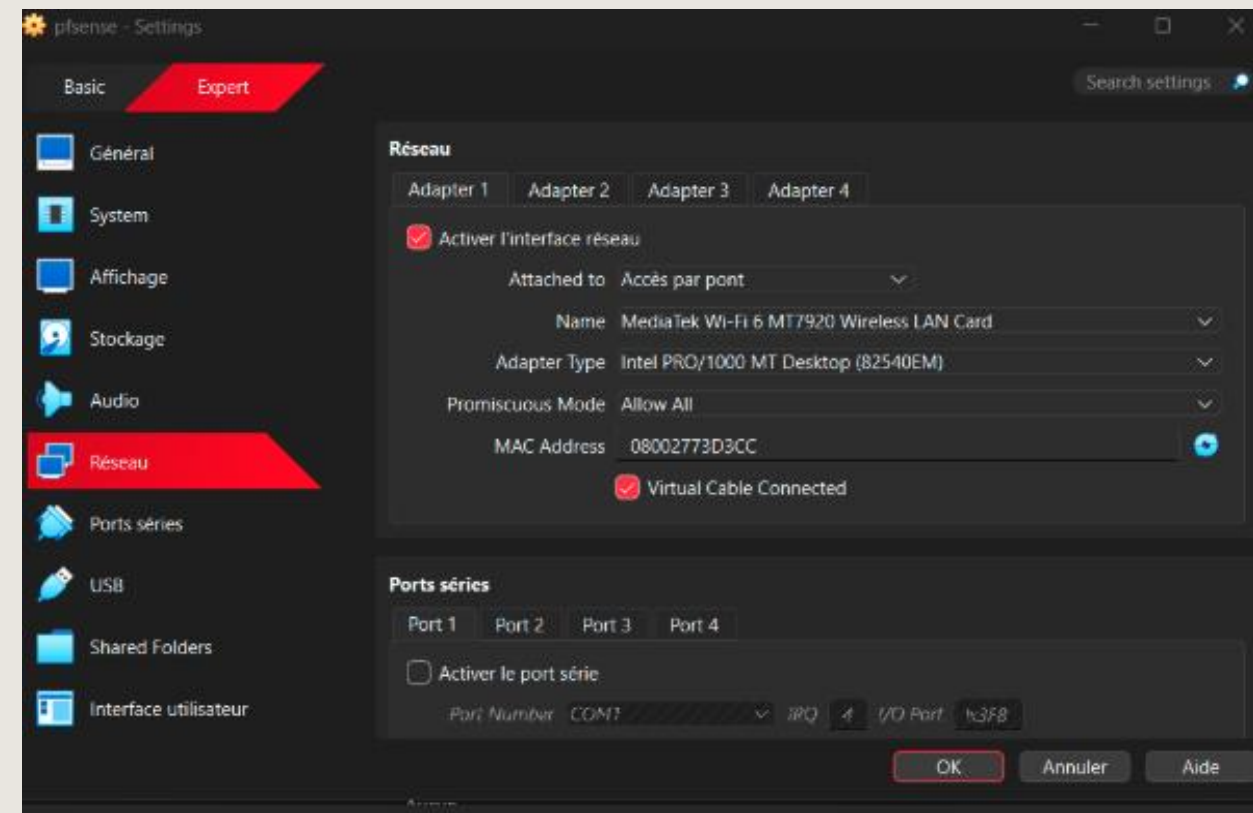
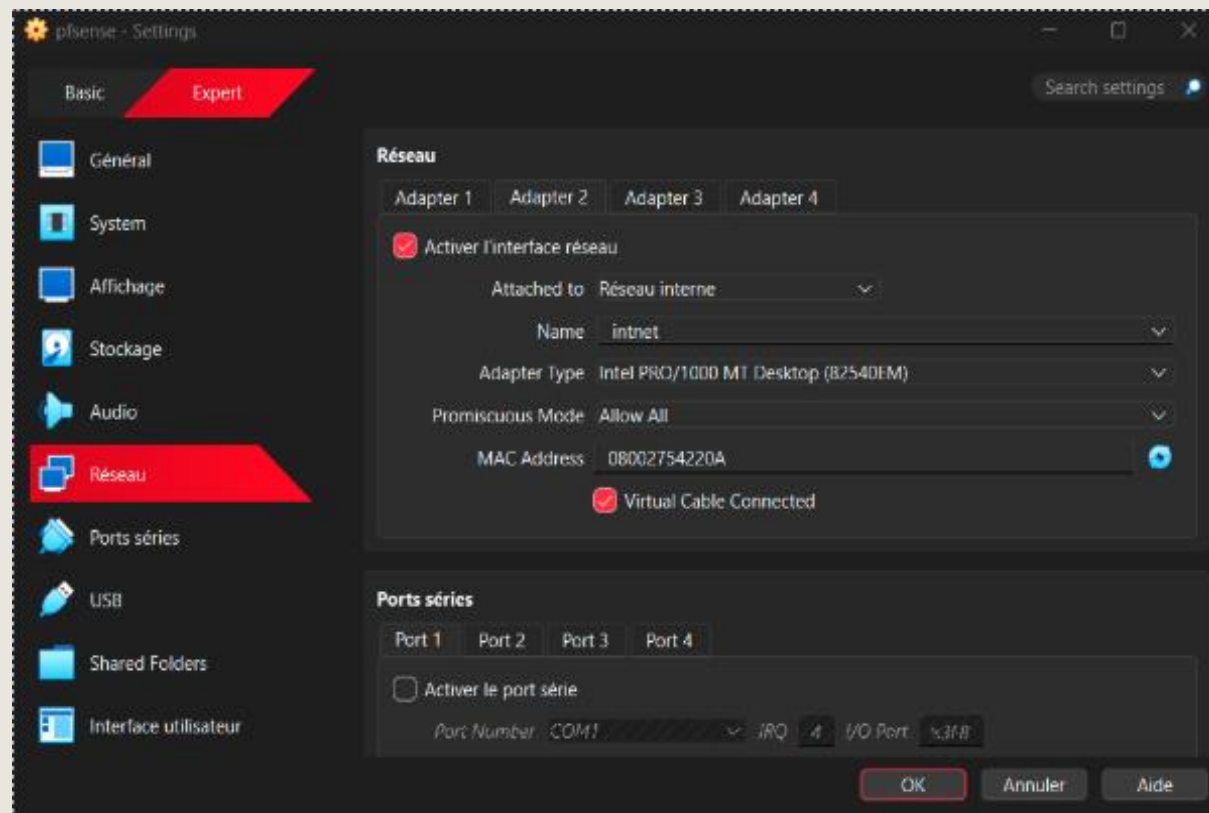
- Réseau - pfSense - LAN 192.168.1.0/24.
- Machines principales SRV-AD-Primary, SRV-AD-Secondary, SRV-FICHIERS et SRV-WAZUH.
- DHCP – Distribution Ip - 192.168.1.100 et 192.168.1.200.

Mise en place des vms

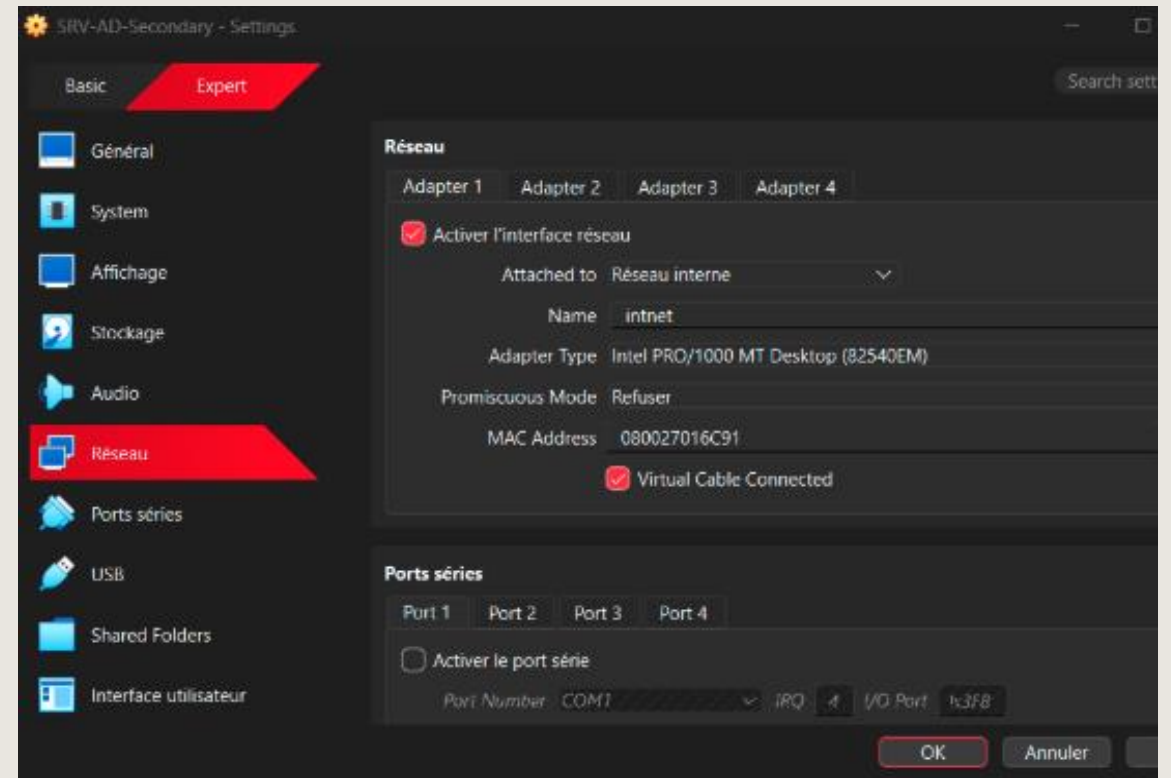
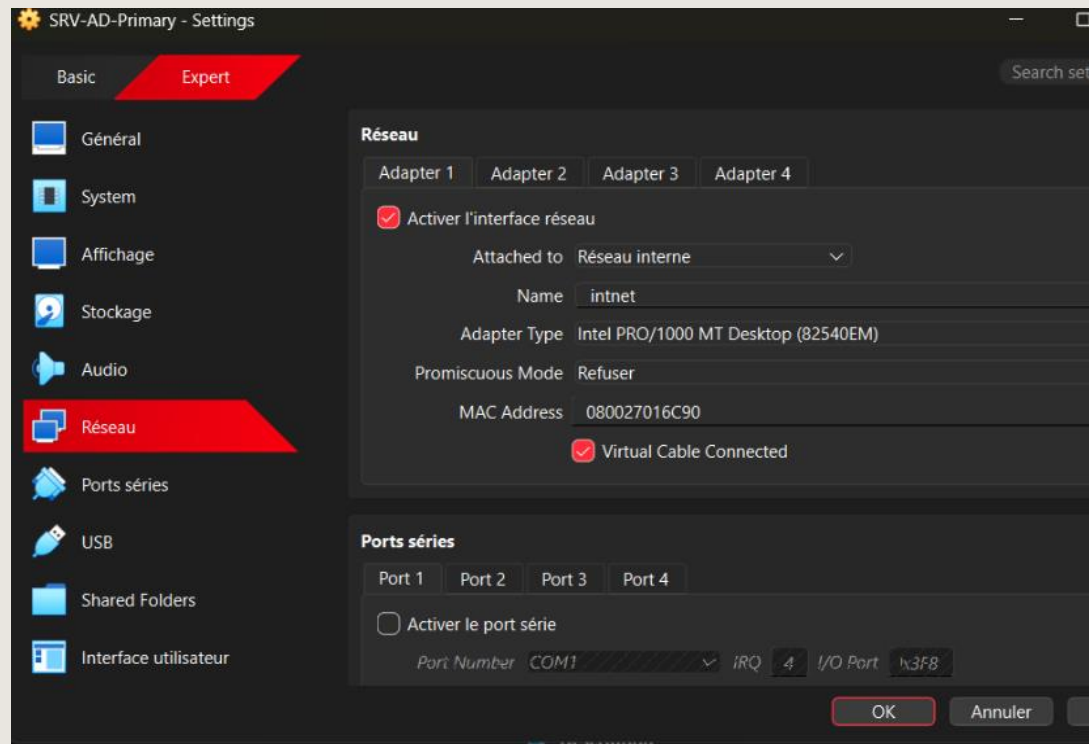
- Environnement virtualisé sous VirtualBox pour simuler l'infrastructure NovaBike.
- pfSense est utilisé comme routeur/pare-feu, et les serveurs Windows sont déployés avec IP fixes.
- Debian 12 a aussi été choisie pour certains services grâce à sa stabilité et sa sécurité.



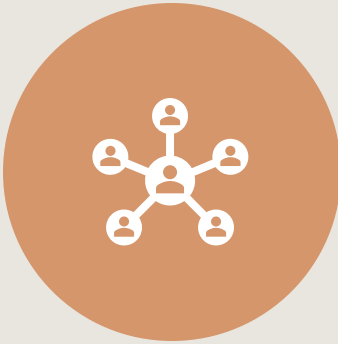
Configuration VM pfSense dans VirtualBox.



Configuration réseau SRV-AD-Primary et Secondary



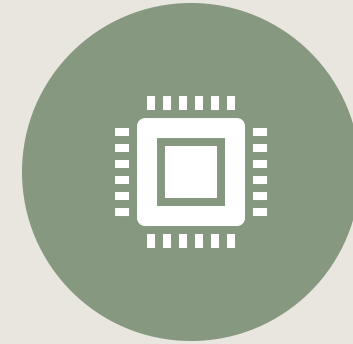
Configuration pfSense



PFSense SÉPARE LE RÉSEAU
INTERNE NOVABIKE D'INTERNET.



L'INTERFACE LAN EST CONFIGURÉE
EN 192.168.1.0/24 AVEC DHCP
TEMPORAIRE AU DÉPART.



LES TESTS MONTRENT UNE
CONNECTIVITÉ FONCTIONNELLE
ENTRE CLIENT, PARE-FEU ET
INTERNET.

```
VirtualBox Virtual Machine - Netgate Device ID: caf86d9ad7be325c9973
*** Welcome to pfSense 2.6.0-RELEASE (amd64) on pfSense ***
WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.20/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24
```

Assignation des interfaces physiques (em0=WAN, em1=LAN)

Test de connectivité depuis un poste client : IP obtenue par DHCP et accès Internet via accès par pont

```
PS C:\Users\vboxuser> ping 192.168.1.1

Envoi d'une requête 'Ping' 192.168.1.1 avec 32 octets de données :
Réponse de 192.168.1.1 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=3 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=2 ms TTL=64

Statistiques Ping pour 192.168.1.1:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 3ms, Moyenne = 2ms
PS C:\Users\vboxuser>
```

Ping pare-feu pfsense

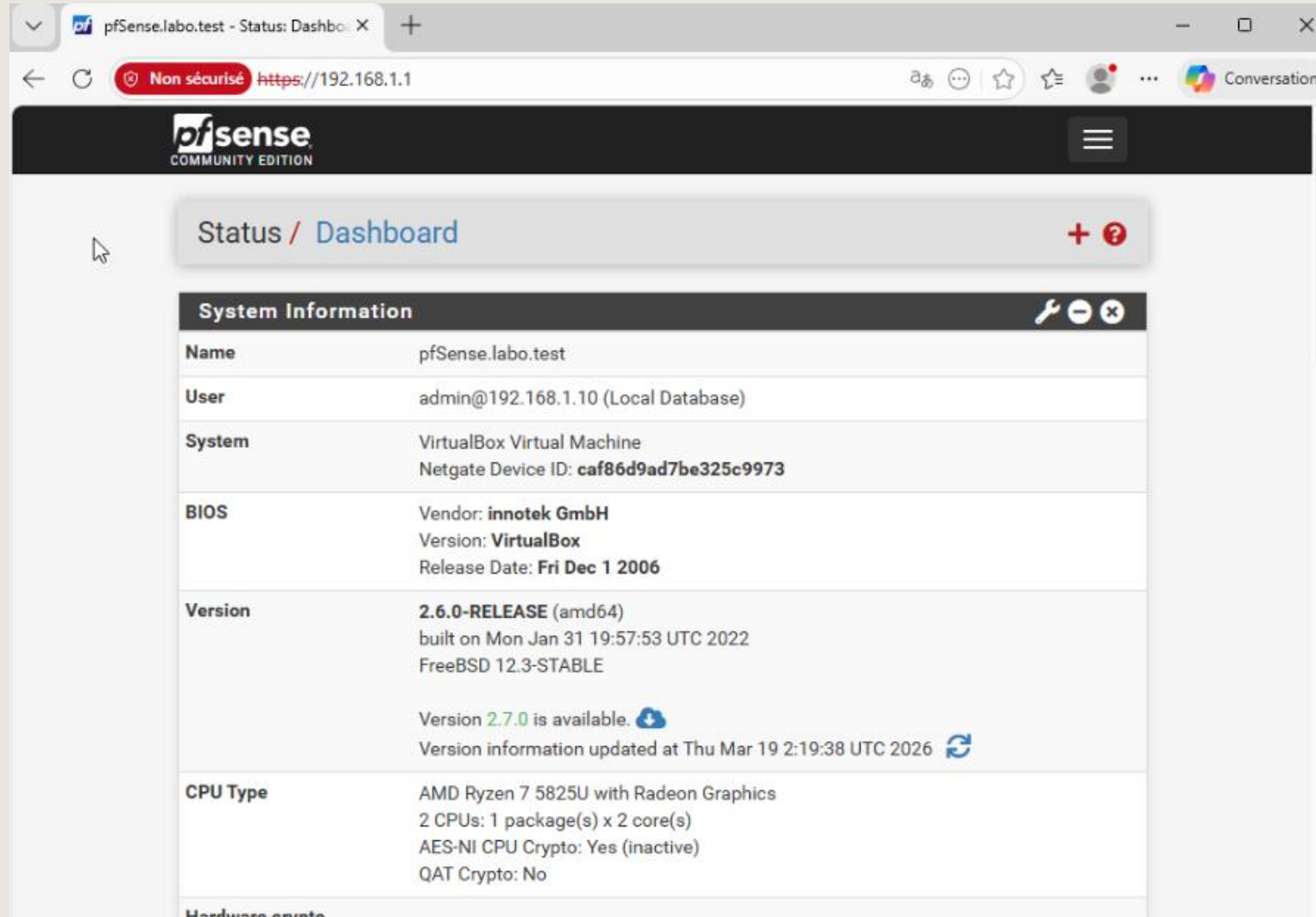
```
PS C:\Users\vboxuser> ping 8.8.8.8

Envoi d'une requête 'Ping' 8.8.8.8 avec 32 octets de données :
Réponse de 8.8.8.8 : octets=32 temps=37 ms TTL=115
Réponse de 8.8.8.8 : octets=32 temps=121 ms TTL=115
Réponse de 8.8.8.8 : octets=32 temps=17 ms TTL=115
Réponse de 8.8.8.8 : octets=32 temps=17 ms TTL=115

Statistiques Ping pour 8.8.8.8:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 17ms, Maximum = 121ms, Moyenne = 48ms
PS C:\Users\vboxuser>
```

Pc client-connexion internet via pare-feu (pfsense)

Connexion à la page pfsense via le pc client réussie



The screenshot shows a web browser window with the address bar displaying 'https://192.168.1.1'. The page title is 'pfSense.labo.test - Status: Dashboard'. The browser's address bar shows a warning 'Non sécurisé' (Not secure) for the https://192.168.1.1 address. The page content is the pfSense Community Edition Status Dashboard. The top navigation bar shows 'Status / Dashboard' with a '+ ?' icon. The main content area is titled 'System Information' and contains a table with the following data:

System Information	
Name	pfSense.labo.test
User	admin@192.168.1.10 (Local Database)
System	VirtualBox Virtual Machine Netgate Device ID: caf86d9ad7be325c9973
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006
Version	2.6.0-RELEASE (amd64) built on Mon Jan 31 19:57:53 UTC 2022 FreeBSD 12.3-STABLE Version 2.7.0 is available.  Version information updated at Thu Mar 19 2:19:38 UTC 2026 
CPU Type	AMD Ryzen 7 5825U with Radeon Graphics 2 CPUs: 1 package(s) x 2 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No

Hardware counts

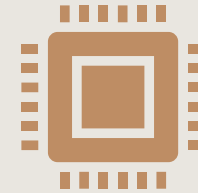
Active Directory, DNS et DHCP



Le domaine **novabike.local** est créé sur le serveur principal avec AD DS et DNS.



Le serveur DHCP distribue les adresses et fournit routeur et DNS aux clients.



Les serveurs Windows Server 2022 sont configurés avec IP statiques et reliés à pfSense.

Afficher/Modifier une relation de basculement
?
X

Modifiez les paramètres relatifs à la relation de basculement :

Nom de la relation : win-kpd2abs9vfg.novabike.local-192.168.1.10

État de ce serveur : Normale

Remplacer par le partenaire hors service

État du serveur partenaire : Normale

☐ Activer l'authentification du message

Secret partagé :

☒ Intervalle de basculement d'état : 15 minutes

Délai de transition maximal du client (MCLT) : 1 heures 0 minutes

☒ Mode d'équilibrage de charge

Serveur local : 60 %
Serveur partenaire : 40 %

☐ Mode du serveur de secours

Rôle de ce serveur : Actif
Adresses réservées pour le serveur de secours : 5 %

OK
Annuler

Config failover DHCP.

```
PS C:\Users\vboxuser> ipconfig /renew

Configuration IP de Windows

Carte Ethernet Ethernet :

    Suffixe DNS propre à la connexion. . . : labo.test
    Adresse IPv6 de liaison locale. . . . . : fe80::314e:e8ea:ad5a:9a12%6
    Adresse IPv4. . . . . : 192.168.1.10
    Masque de sous-réseau. . . . . : 255.255.255.0
    Passerelle par défaut. . . . . : fe80::a00:27ff:fe54:220a%6
                                     192.168.1.1

PS C:\Users\vboxuser>
```

Adresse pc connecté au serveur DHCP principal

```
PS C:\Users\vboxuser> ipconfig /renew

Configuration IP de Windows

Carte Ethernet Ethernet :

    Suffixe DNS propre à la connexion. . . : labo.test
    Adresse IPv6 de liaison locale. . . . . : fe80::314e:e8ea:ad5a:9a12%6
    Adresse IPv4. . . . . : 192.168.1.10
    Masque de sous-réseau. . . . . : 255.255.255.0
    Passerelle par défaut. . . . . : fe80::a00:27ff:fe54:220a%6
                                     192.168.1.1
```

Le pc client
n'a plus d'adresse ip quand on a éteint le
serveur DHCP principal

```
PS C:\Users\vboxuser> ipconfig /renew
```

```
Configuration IP de Windows
```

```
Carte Ethernet Ethernet :
```

```
Suffixe DNS propre à la connexion. . . : labo.test
Adresse IPv6 de liaison locale. . . . . : fe80::314e:e8ea:ad5a:9a12%6
Adresse IPv4. . . . . : 192.168.1.10
Masque de sous-réseau. . . . . : 255.255.255.0
Passerelle par défaut. . . . . : fe80::a00:27ff:fe54:220a%6
                                   192.168.1.1
```

Quand on fait un ipconfig /renew le pc obtient une adresse ip donnée par le serveur dhcp de redondance

```
PS C:\Users\vboxuser> ping 192.168.1.1
```

```
Envoi d'une requête 'Ping' 192.168.1.1 avec 32 octets de données :
```

```
Réponse de 192.168.1.1 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=2 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=1 ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps=1 ms TTL=64
```

```
Statistiques Ping pour 192.168.1.1:
```

```
Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
Minimum = 1ms, Maximum = 2ms, Moyenne = 1ms
```

Et le pc arrive à ping le pare-feu

Utilisateurs et groupes Active Directory

- Groupes de sécurité par service créés dans l'UO NovaBike.
- Comptes utilisateurs créés et rattachés aux groupes métiers pour simplifier les droits.
- Cette organisation permet une gestion plus propre et plus sécurisée des accès.

Script de creation de groupes et utilisateurs dans l'AD

```
powershell

# 1. Créer OU
New-ADOrganizationalUnit -Name "Utilisateurs NovaBike" -Path "DC=novabike,DC=local"
-ProtectedFromAccidentalDeletion $true

# 2. Utilisateurs réalistes
$Users = @(
    @{Name="Gerard
NOVAKI";Sam="gerant.novabike";UPN="gerant.novabike@novabike.local";Pass="GerardVelo
2026!";Desc="Gérant"},
    @{Name="Alain
TAVERNE";Sam="admin.atelier";UPN="admin.atelier@novabike.local";Pass="AlainAtelier2
026!";Desc="Chef atelier"},
    @{Name="Beatrice
LECOIN";Sam="admin.boutique";UPN="admin.boutique@novabike.local";Pass="BeaBoutique2
026!";Desc="Responsable boutique"},
    @{Name="Cedric
MARCHAND";Sam="commercial";UPN="commercial@novabike.local";Pass="CedricCom2026!";De
sc="Commercial itinérant"}
)

foreach ($Su in $Users) {
    New-ADUser -Name $Su.Name -SamAccountName $Su.Sam -UserPrincipalName $Su.UPN -
AccountPassword (ConvertTo-SecureString $Su.Pass -AsPlainText -Force) -Enabled $true
-PasswordNeverExpires $true -Description $Su.Desc -Path "OU=Utilisateurs
NovaBike,DC=novabike,DC=local" -ProtectedFromAccidentalDeletion $true
    Write-Host "✅ Créé : $($Su.Sam)"
}

# 3. Groupes sécurité
$SGroups = @("G-ATELIER","G-BOUTIQUE","G-COMMERCIAL","G-GERANTS")
foreach ($Sg in $SGroups) {
    New-ADGroup -Name $Sg -GroupScope Global -GroupCategory Security -Path
"OU=Utilisateurs NovaBike,DC=novabike,DC=local" -Description "Groupe sécurité $Sg" -
ProtectedFromAccidentalDeletion $true
```

```
}

# 4. Associer membres
Add-ADGroupMember "G-ATELIER" "admin.atelier"
Add-ADGroupMember "G-BOUTIQUE" "admin.boutique"
Add-ADGroupMember "G-COMMERCIAL" "commercial"
Add-ADGroupMember "G-GERANTS" @("gerant.novabike","administrateur")

Write-Host "| TOUS utilisateurs et groupes créés ! Vérifie dsa.msc"
```



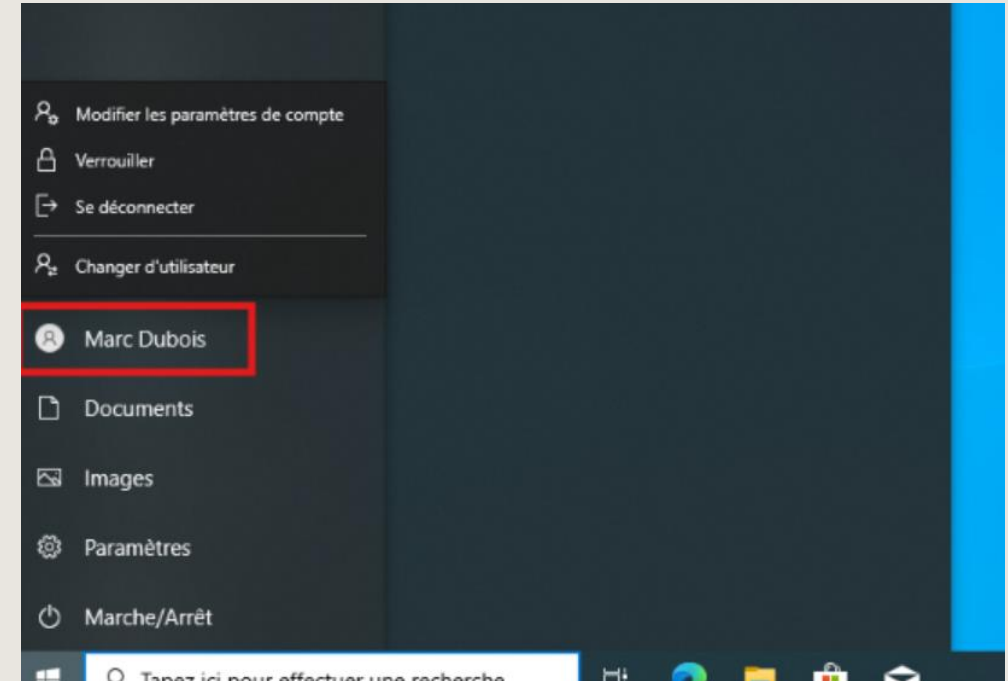
```
PS C:\Users\Administrateur> Get-ADUser -Filter * -SearchBase "OU=Utilisateurs NovaBike,DC=novabike,DC=local"

DistinguishedName : CN=Gerard NOVAKI,OU=Utilisateurs NovaBike,DC=novabike,DC=local
Enabled           : True
GivenName        :
Name             : Gerard NOVAKI
ObjectClass       : user
ObjectGUID        : 7bab7b50-181a-4edb-a0ca-0490f8a8f487
SamAccountName    : gerant.novabike
SID              : S-1-5-21-468841071-3659426184-2653338209-1110
Surname          :
UserPrincipalName : gerant.novabike@novabike.local

DistinguishedName : CN=Alain TAVERNE,OU=Utilisateurs NovaBike,DC=novabike,DC=local
Enabled           : True
GivenName        :
Name             : Alain TAVERNE
ObjectClass       : user
ObjectGUID        : 94c0bf4d-29d4-4ef6-bf61-6489dc036fe5
SamAccountName    : admin.atelier
SID              : S-1-5-21-468841071-3659426184-2653338209-1111
Surname          :
UserPrincipalName : admin.atelier@novabike.local

DistinguishedName : CN=Beatrice LECOIN,OU=Utilisateurs NovaBike,DC=novabike,DC=local
Enabled           : True
GivenName        :
Name             : Beatrice LECOIN
ObjectClass       : user
ObjectGUID        : cadb4622-f914-4ef7-9360-8430c3870c3c
SamAccountName    : admin.boutique
SID              : S-1-5-21-468841071-3659426184-2653338209-1112
Surname          :
UserPrincipalName : admin.boutique@novabike.local
```

Comptes utilisateurs NovaBike créés et actifs dans Active Directory.



Test connexion avec un utilisateur

Les règles du gpo émis



La taille minimum du mot de passe : 8.



Le mot de pass doit Etre complexe : Activé.



Maximum de validité du mot de passe : 60 jours.



Nombre de tentative : 5.

Serveur de fichiers et FTP

- Un serveur de fichiers sécurisé a été mis en place avec partage SMB.
- Les droits NTFS et les accès sont gérés par groupes Active Directory.
- Un site FTP IIS a aussi été configuré pour exposer les fichiers via FTP dans le labo.

```
PS C:\Users\Administrateur> New-NetFirewallRule -DisplayName "Autoriser Ping ICMPv4" -Direction Inbound -Protocol ICMPv4 -IcmpType 8 -Action Allow

Name                           : {1493db66-e9e4-42fa-9f4c-3e0da891dc30}
DisplayName                     : Autoriser Ping ICMPv4
Description                     :
DisplayGroup                    :
Group                           :
Enabled                         : True
Profile                         : Any
Platform                       : {}
Direction                      : Inbound
Action                         : Allow
EdgeTraversalPolicy             : Block
LooseSourceMapping              : False
LocalOnlyMapping               : False
Owner                           :
PrimaryStatus                   : OK
Status                         : La règle a été analysée à partir de la banque. (65536)
EnforcementStatus               : NotApplicable
PolicyStoreSource               : PersistentStore
PolicyStoreSourceType           : Local
RemoteDynamicKeywordAddresses  : {}

PS C:\Users\Administrateur> ping 192.168.1.11

Envoi d'une requête 'Ping' 192.168.1.11 avec 32 octets de données :
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.

Statistiques Ping pour 192.168.1.11:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
PS C:\Users\Administrateur> _
```

SRV-FICHIERS joint au domaine.

Règle ICMP rétablie / diagnostic réseau résolu.

```
PS C:\Users\Administrateur> New-NetFirewallRule -DisplayName "Autoriser Ping ICMPv4" -Direction Inbound -Protocol ICMPv4 -IcmpType 8 -Action Allow

Name
-----
DisplayName      : Autoriser Ping ICMPv4
Description      :
DisplayGroup     :
Group            :
Enabled          : True
Profile          : Any
Platform         : {}
Direction        : Inbound
Action           : Allow
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping  : False
Owner            :
PrimaryStatus    : OK
Status           : La règle a été analysée à partir de la banque. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses : {}

PS C:\Users\Administrateur> ping 192.168.1.11

Envoi d'une requête 'Ping' 192.168.1.11 avec 32 octets de données :
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.
Réponse de 192.168.1.15 : Impossible de joindre l'hôte de destination.

Statistiques Ping pour 192.168.1.11:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
PS C:\Users\Administrateur> _
```

Diagnostic résolu : après création règle "Autoriser Ping ICMPv4", connectivité réseau SRV-AD-Primary (192.168.1.11) rétablie.

Get-SmbShare ou partage “Commun”.

```
PS C:\Users\Administrateur> New-Item -Path "C:\Partage_Commune" -ItemType Directory
```

Répertoire : C:\

Mode	LastWriteTime	Length	Name
d-----	03/04/2026 14:53		Partage_Commune

Propriétés de sécurité du dossier avec groupes AD.

Autorisations pour Partage_Comm

Sécurité

Nom de l'objet : C:\Partage_Comm

Noms de groupes ou d'utilisateurs :

- CREATEUR PROPRIETAIRE
- Système
- GG_Direction (NOVABIKE\GG_Direction)
- Administrateurs (NOVABIKE\Administrateurs)
- GG_Technique (NOVABIKE\GG_Technique)
- Utilisateurs (NOVABIKE\Utilisateurs)

Ajouter... Supprimer

Autorisations pour GG_Technique

	Autoriser	Refuser
Lecture et exécution	☒	☐
Affichage du contenu du dossier	☒	☐
Lecture	☒	☐
Écriture	☒	☐
Autorisations spéciales	☐	☐

[Informations sur le contrôle d'accès et les autorisations](#)

OK Annuler Appliquer

Sécurité

Nom de l'objet : C:\Partage_Comm

Noms de groupes ou d'utilisateurs :

- CREATEUR PROPRIETAIRE
- Système
- Administrateurs (NOVABIKE\Administrateurs)
- GG_Direction (NOVABIKE\GG_Direction)
- Utilisateurs (NOVABIKE\Utilisateurs)

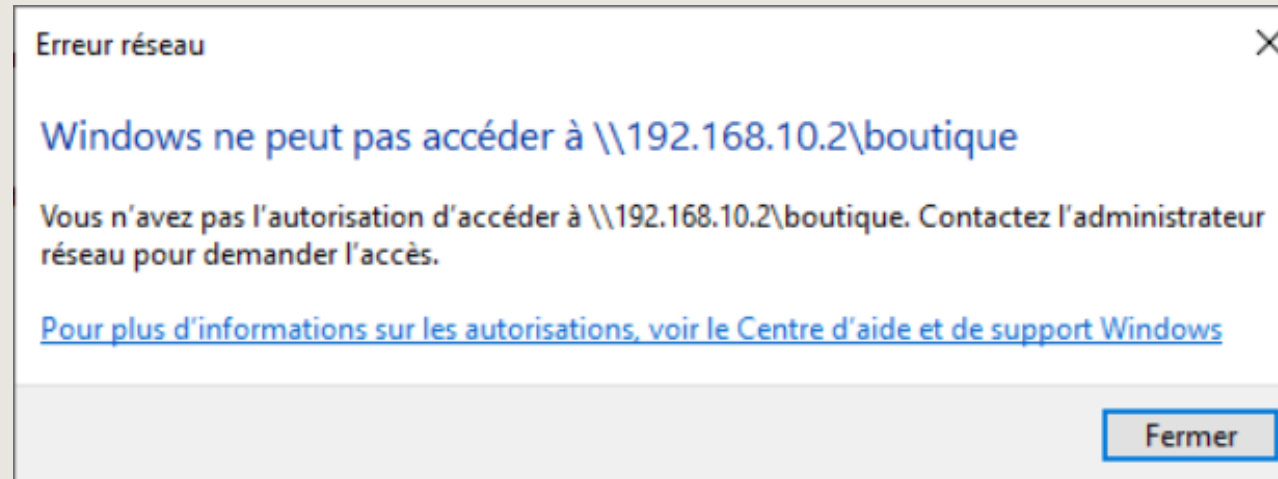
Ajouter... Supprimer

Autorisations pour GG_Direction

	Autoriser	Refuser
Lecture et exécution	☒	☐
Affichage du contenu du dossier	☒	☐
Lecture	☒	☐
Écriture	☒	☐
Autorisations spéciales	☐	☐

[Informations sur le contrôle d'accès et les autorisations](#)

OK Annuler Appliquer



Compte Emma :

Tentative d'accès à tous les dossiers

Accès refusé sauf pour ADMINISTRATION

Création de dossier possible dans ADMINISTRATION

Conclusion :

Les permissions sont correctement configurées pour Emma.



Compte Thomas Durand (Maintenance) :

Accès autorisé au dossier MAINTENANCE

Création de fichier possible

Accès refusé au dossier ADMINISTRATION

Conclusion :

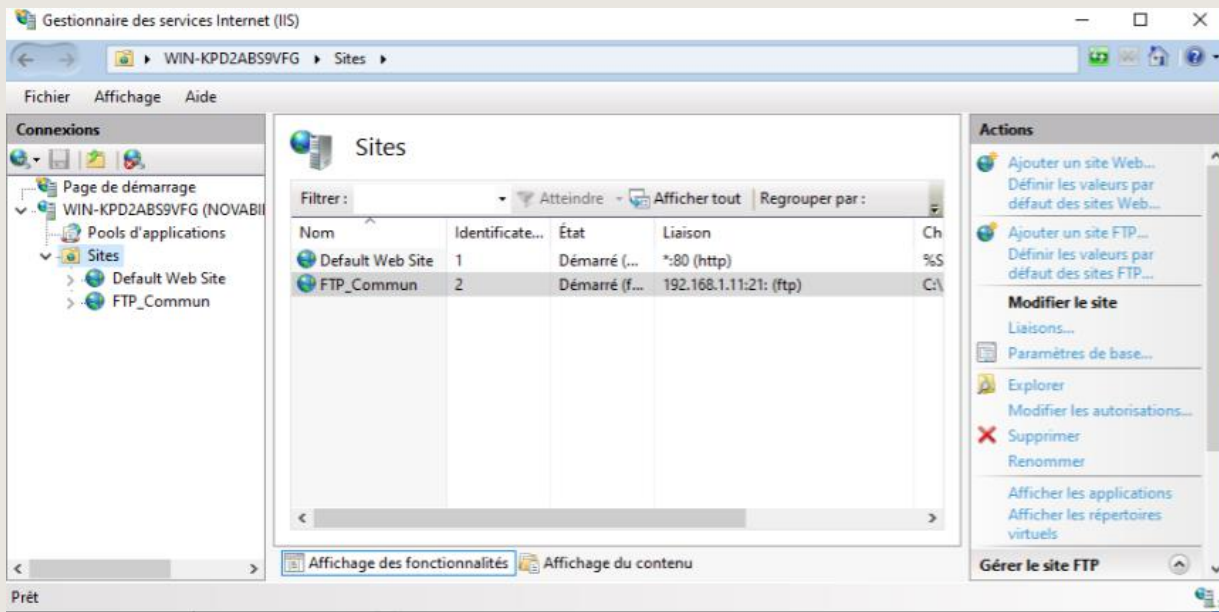
Les droits d'accès sont respectés selon les groupes utilisateurs.

IIS Manager avec FTP_Commune.

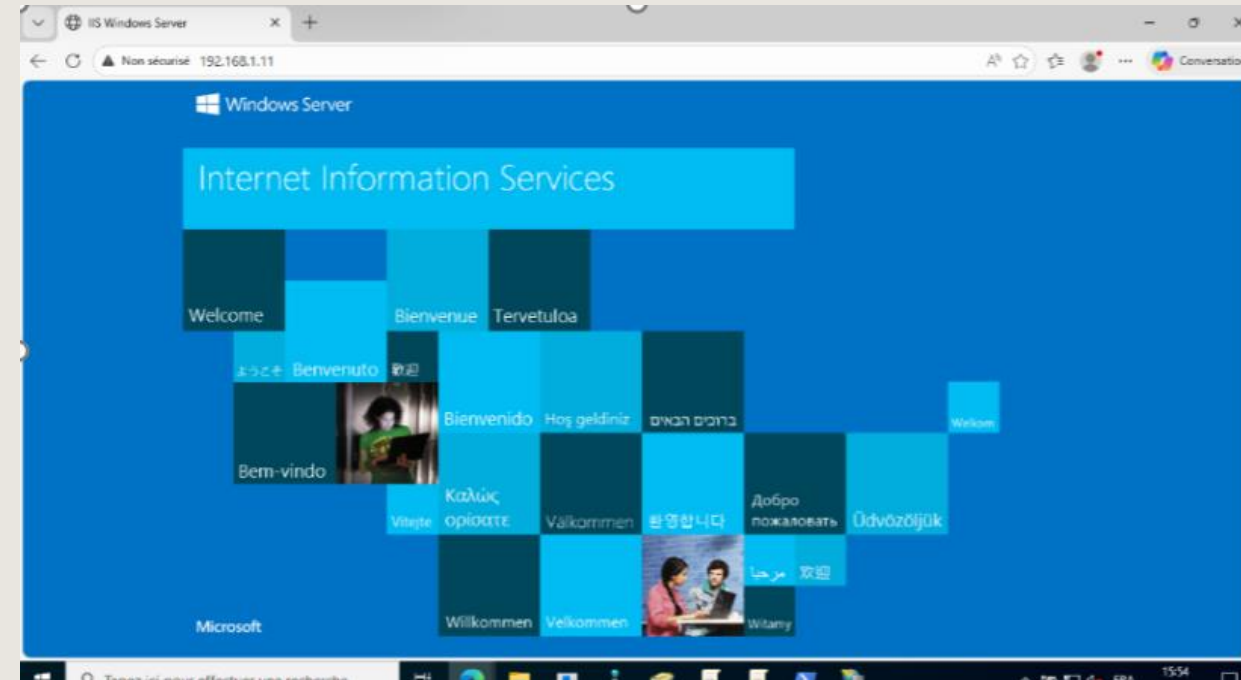
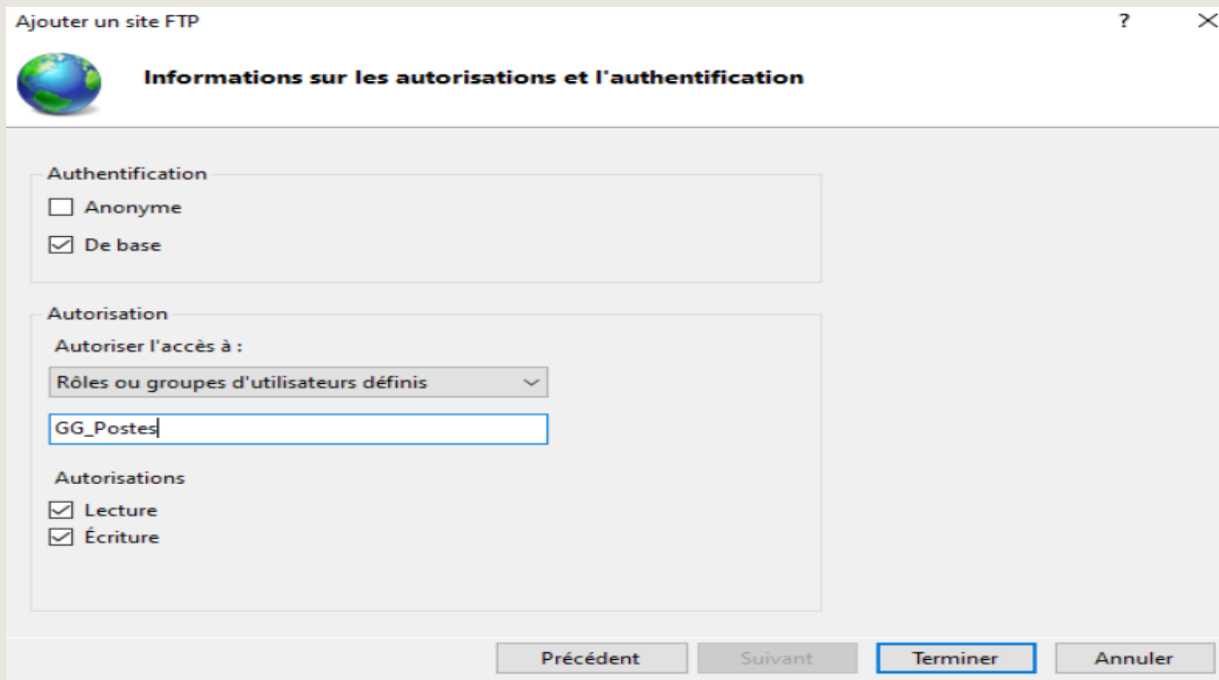
```
PS C:\Users\Administrateur> Install-WindowsFeature Web-Server, Web-FTP-Server -IncludeManagementTools
```

Success	Restart Needed	Exit Code	Feature Result
True	No	Success	{Fonctionnalités HTTP communes, Document p...

IIS + FTP installé sur SRV-AD_1



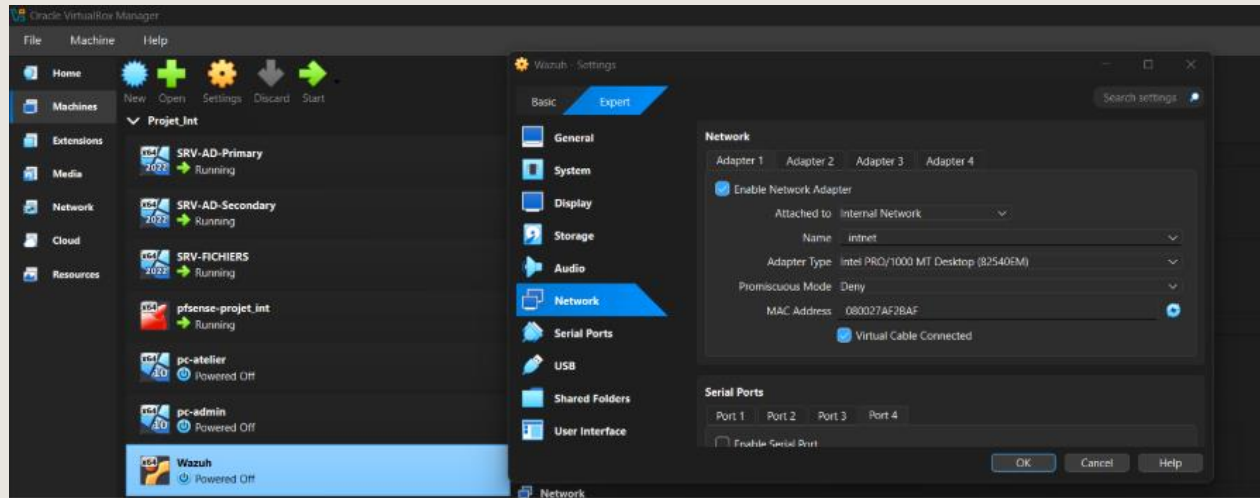
IIS Manager avec FTP_Commun dans Sites



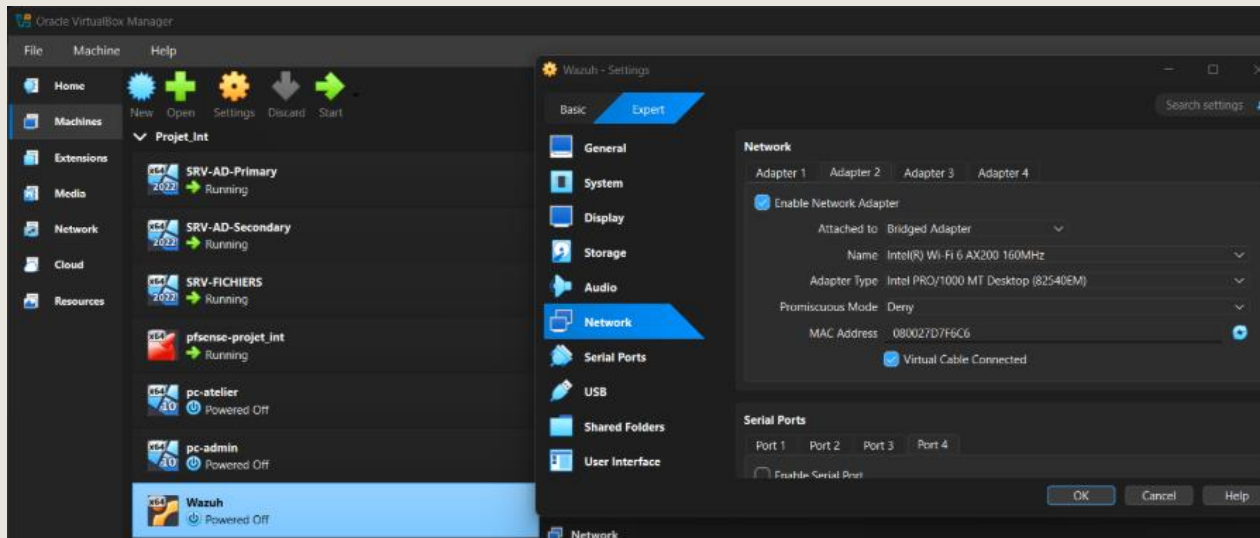
Fenêtre de configuration du site FTP.



Configuration - gestionnaire wazuh



Carte réseau 1 : Réseau interne



Carte réseau 2 : Mode pont

```
wazuhmanager@wazuhmanager: ~
ssh_dispatch_run_fatal: Connection to 192.168.1.104 port 22: Unknown error
PS C:\Users\Administrateur> ssh wazuhmanager@192.168.1.104
wazuhmanager@192.168.1.104's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Apr 22 11:26:15 AM UTC 2026

System load:  0.01          Processes:           106
Usage of /:   23.6% of 11.21GB Users logged in:       1
Memory usage: 1%           IPv4 address for enp0s3: 192.168.1.104
Swap usage:   0%           IPv4 address for enp0s8: 192.168.0.38

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

wazuhmanager@wazuhmanager: ~$
```

Connexion gestionnaire wazuh via ssh

```
root@wazuhmanager: /home/wazuhmanager
wazuhmanager@wazuhmanager:~$ apt-get install gnupg apt-transport-https
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
wazuhmanager@wazuhmanager:~$ sudo su
[sudo] password for wazuhmanager:
Sorry, try again.
[sudo] password for wazuhmanager:
root@wazuhmanager:/home/wazuhmanager# apt-get install gnupg apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gnupg is already the newest version (2.4.4-2ubuntu17.4).
gnupg set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 3,970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3,970 B]
Fetched 3,970 B in 0s (29.6 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 88019 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Setting up apt-transport-https (2.8.3) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuhmanager:/home/wazuhmanager#
```

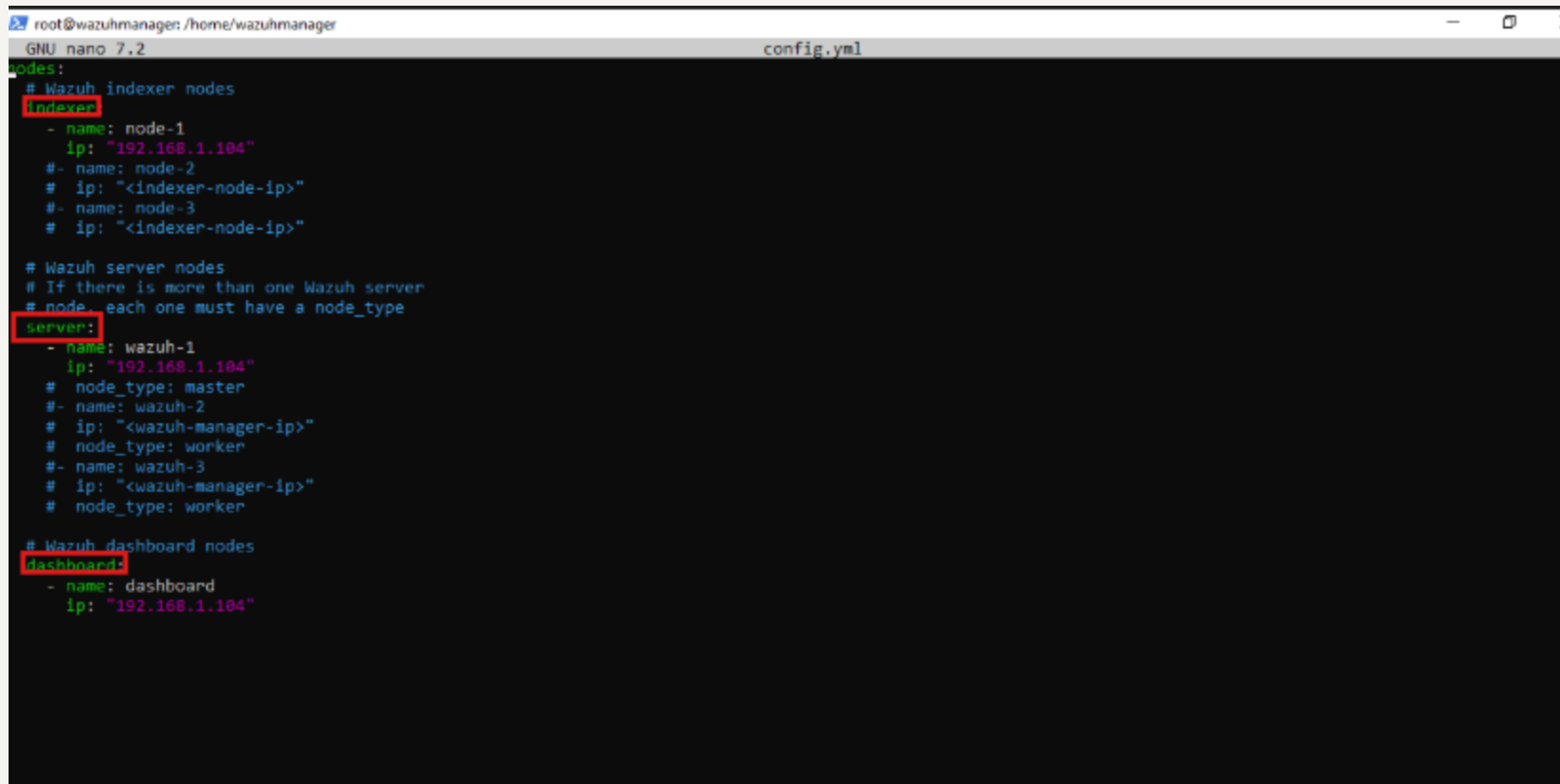
Installation paquets nécessaires.

Téléchargement du script d'installation :

curl -sO <https://packages.wazuh.com/4.7/wazuh-install.sh>

Fichier de configuration

curl -sO <https://packages.wazuh.com/4.7/config.yml>



```
root@wazuhmanager: /home/wazuhmanager
GNU nano 7.2 config.yml
nodes:
# Wazuh indexer nodes
indexer:
  - name: node-1
    ip: "192.168.1.104"
  #- name: node-2
  # ip: "<indexer-node-ip>"
  #- name: node-3
  # ip: "<indexer-node-ip>"

# Wazuh server nodes
# If there is more than one Wazuh server
# node_ each one must have a node_type
server:
  - name: wazuh-1
    ip: "192.168.1.104"
    # node_type: master
  #- name: wazuh-2
  # ip: "<wazuh-manager-ip>"
  # node_type: worker
  #- name: wazuh-3
  # ip: "<wazuh-manager-ip>"
  # node_type: worker

# Wazuh dashboard nodes
dashboard:
  - name: dashboard
    ip: "192.168.1.104"
```

nano config.yml

Installation wazuh

bash wazuh-install.sh -a

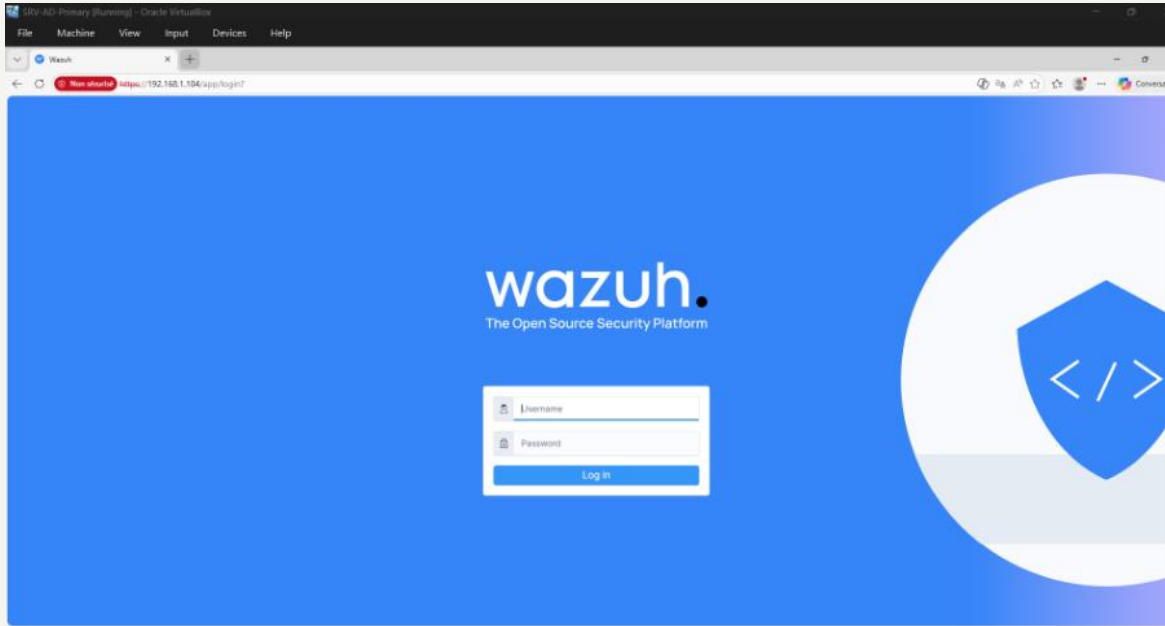
```
root@wazuhmanager:/home/wazuhmanager# bash wazuh-install.sh -a
22/04/2020 11:50:35 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
22/04/2020 11:50:35 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/04/2020 11:50:35 ERROR: The recommended systems are: Red Hat Enterprise Linux 7, 8, 9; CentOS 7, 8; Amazon Linux 2; Ubuntu 18.04, 19.04, 20.04, 22.04. The current system does not match this list. Use -i|--ignore-check to skip this check.
root@wazuhmanager:/home/wazuhmanager# bash wazuh-install.sh -a -i
22/04/2020 11:52:28 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
22/04/2020 11:52:28 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/04/2020 11:52:51 WARNING: Hardware and system checks ignored.
22/04/2020 11:52:51 INFO: Wazuh web interface port will be 443.
22/04/2020 11:53:05 INFO: Wazuh repository added.
22/04/2020 11:53:05 INFO: --- Configuration files ---
22/04/2020 11:53:05 INFO: Generating configuration files.
22/04/2020 11:53:08 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
22/04/2020 11:53:09 INFO: --- Wazuh indexer ---
22/04/2020 11:53:09 INFO: Starting Wazuh indexer installation.
```

```
root@wazuhmanager:/home/wazuhmanager# bash wazuh-install.sh -a -i
22/04/2020 11:52:28 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
22/04/2020 11:52:28 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/04/2020 11:52:51 WARNING: Hardware and system checks ignored.
22/04/2020 11:52:51 INFO: Wazuh web interface port will be 443.
22/04/2020 11:53:05 INFO: Wazuh repository added.
22/04/2020 11:53:05 INFO: --- Configuration files ---
22/04/2020 11:53:05 INFO: Generating configuration files.
22/04/2020 11:53:08 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
22/04/2020 11:53:09 INFO: --- Wazuh indexer ---
22/04/2020 11:53:09 INFO: Starting Wazuh indexer installation.
22/04/2020 11:53:20 INFO: Wazuh indexer installation finished.
22/04/2020 11:53:20 INFO: Wazuh indexer post install configuration finished.
22/04/2020 11:53:20 INFO: Starting service wazuh-indexer.
22/04/2020 11:56:48 INFO: wazuh-indexer service started.
22/04/2020 11:56:48 INFO: Initializing Wazuh indexer cluster security settings.
22/04/2020 11:57:01 INFO: Wazuh indexer cluster initialized.
22/04/2020 11:57:01 INFO: Wazuh server
22/04/2020 11:57:01 INFO: Starting the Wazuh manager installation.
22/04/2020 12:00:23 INFO: Wazuh manager installation finished.
22/04/2020 12:00:23 INFO: Starting service wazuh-manager.
22/04/2020 12:00:45 INFO: wazuh-manager service started.
22/04/2020 12:00:45 INFO: Starting Filebeat installation.
22/04/2020 12:01:07 INFO: Filebeat installation finished.
22/04/2020 12:01:09 INFO: Filebeat post install configuration finished.
22/04/2020 12:01:09 INFO: Starting service filebeat.
22/04/2020 12:01:14 INFO: Filebeat service started.
22/04/2020 12:01:14 INFO: Wazuh dashboard
22/04/2020 12:01:14 INFO: Starting Wazuh dashboard installation.
22/04/2020 12:04:40 INFO: Wazuh dashboard installation finished.
22/04/2020 12:04:40 INFO: Wazuh dashboard post install configuration finished.
22/04/2020 12:04:49 INFO: Starting service wazuh-dashboard.
22/04/2020 12:04:50 INFO: wazuh-dashboard service started.
22/04/2020 12:05:47 INFO: Initializing Wazuh dashboard web application.
22/04/2020 12:05:48 INFO: Wazuh dashboard web application initialized.
22/04/2020 12:05:48 INFO: --- Summary ---
22/04/2020 12:05:48 INFO: You can access the web interface https://wazuh-dashboard-ip:443
User: admin
Password: TSvY2KF5HwB0wHT.L78dQdEKipex6y3
22/04/2020 12:05:48 INFO: Installation finished.
root@wazuhmanager:/home/wazuhmanager#
```

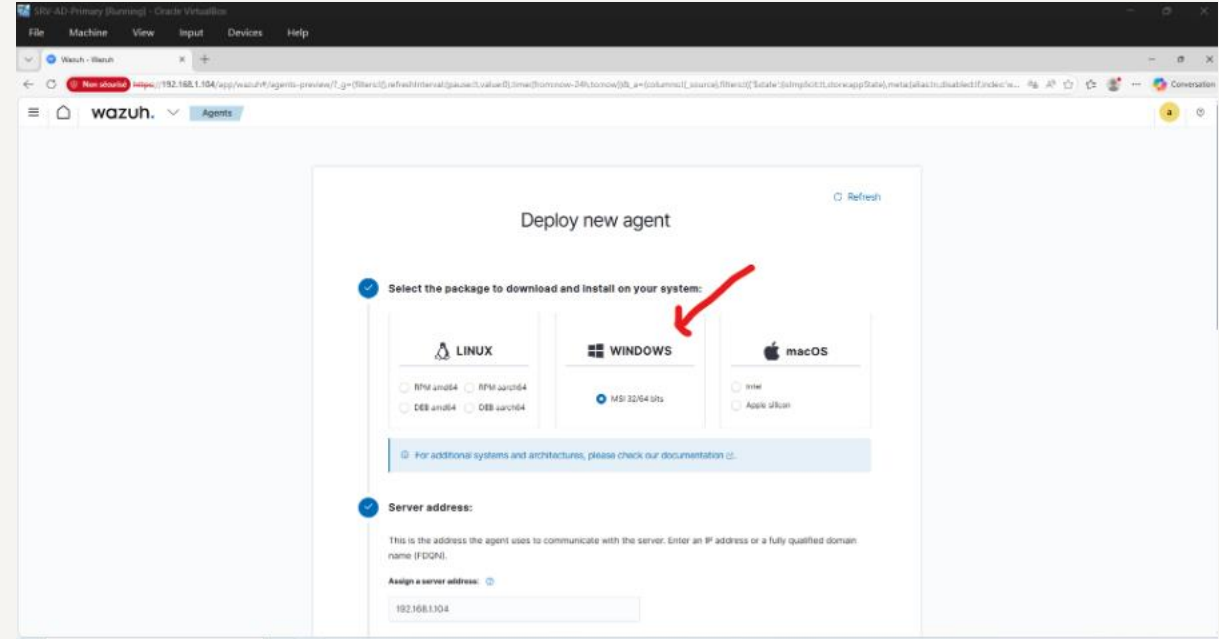
Programme d'installation terminé

Accéder au Dashboard

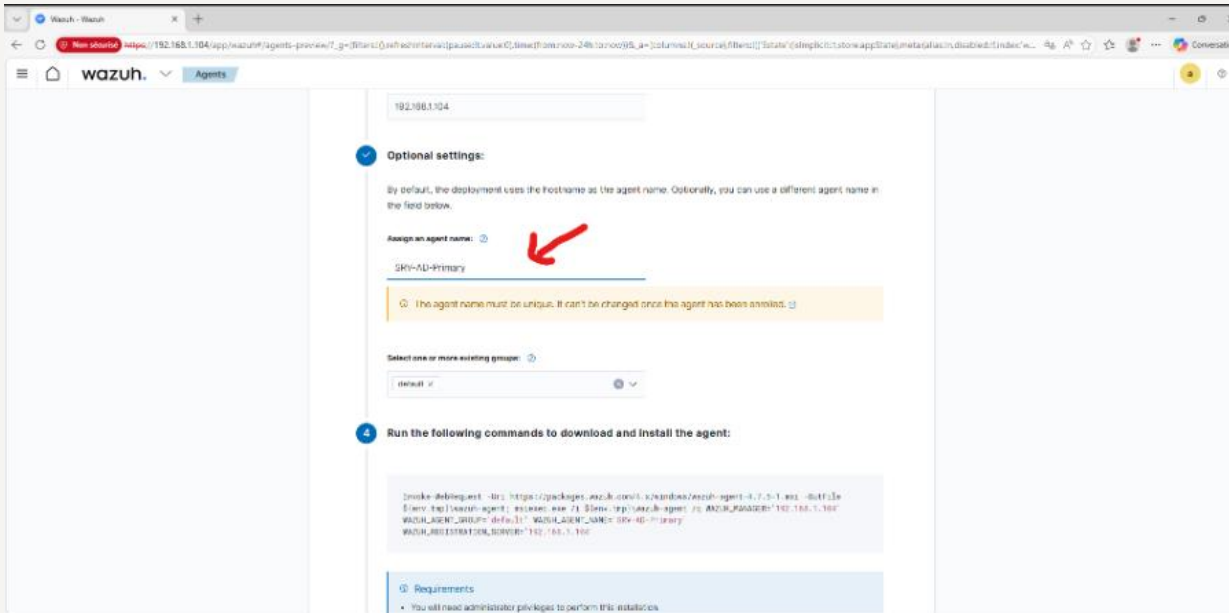
Navigateur sur PC Windows :
<https://192.168.1.104>



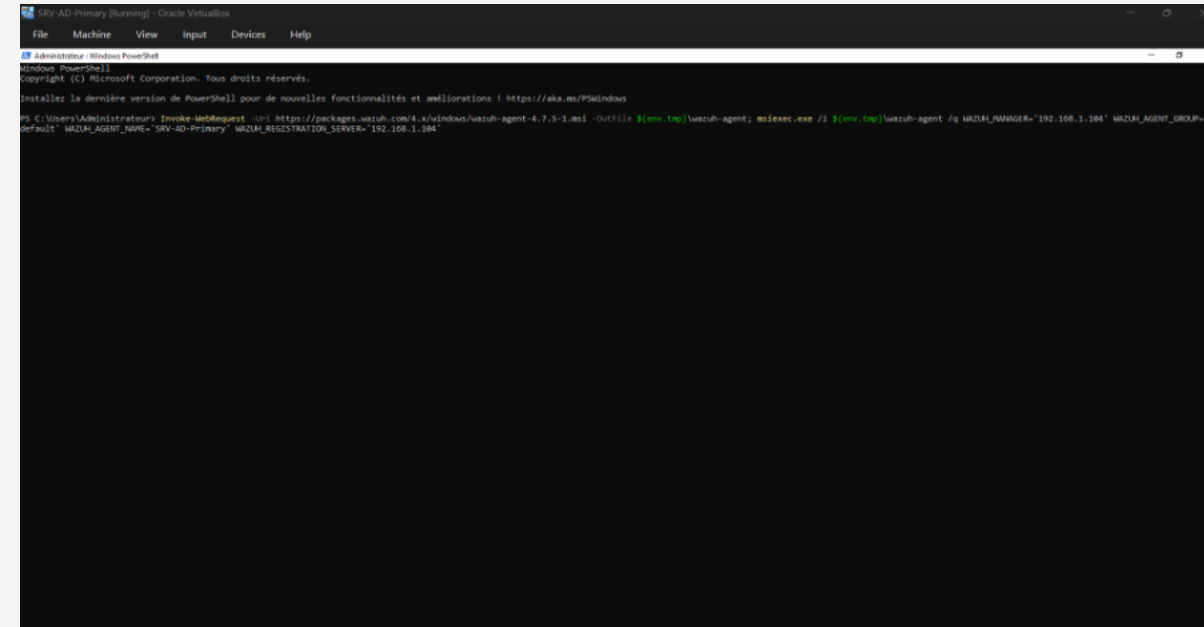
Authentication



Déploiement



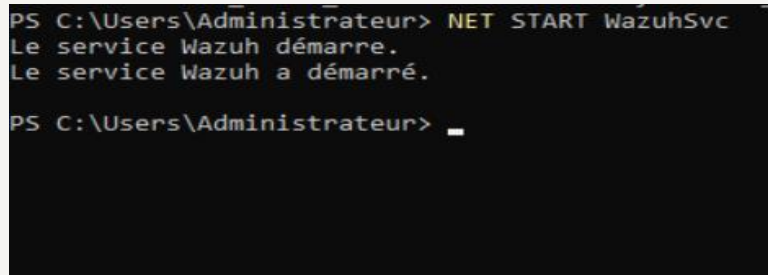
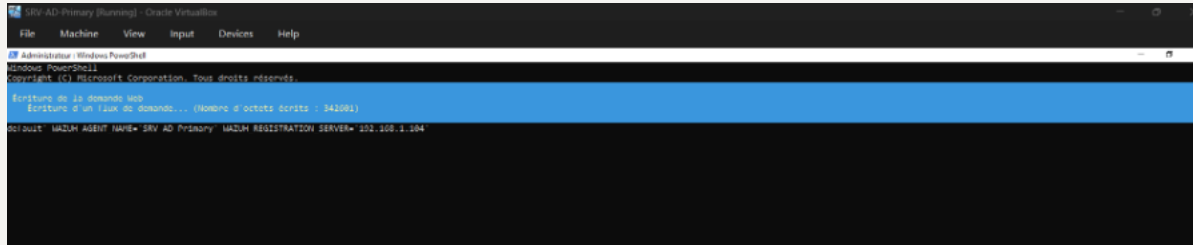
Installation agent via Cli



Commande utilisée:

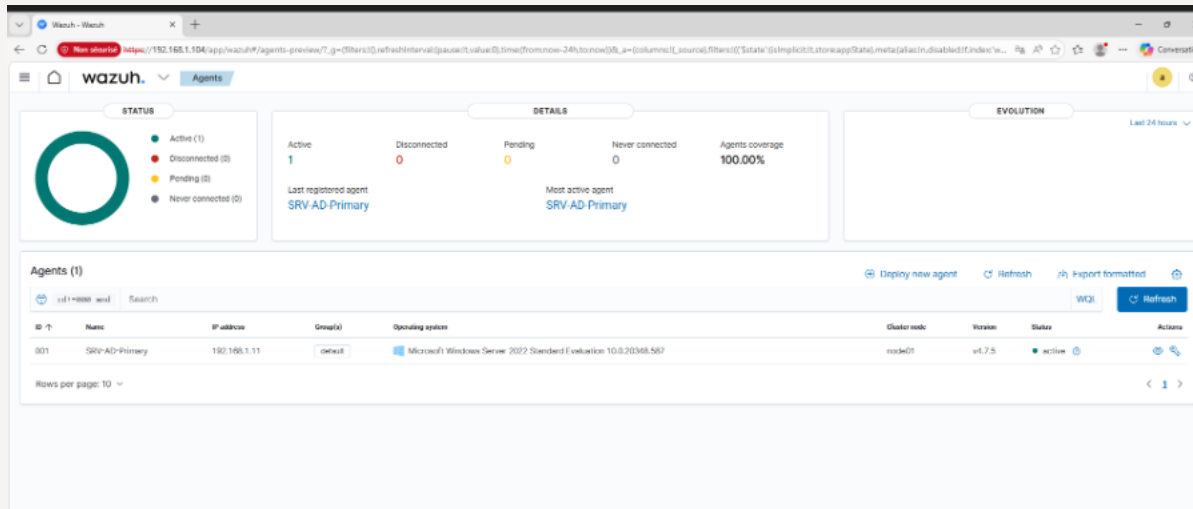
```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.5-1.msi -  
OutFile ${{env.tmp}}\wazuh-agent; msisexec.exe /i ${{env.tmp}}\wazuh-agent /q  
WAZUH_MANAGER=192.168.1.104 WAZUH_AGENT_GROUP=default  
WAZUH_AGENT_NAME=SRV-AD-Primary WAZUH_REGISTRATION_SERVER=192.168.1.104'
```


Démarrage agent depuis Cli



NET START WazuhSvc

AD-Primary - Agent démarré



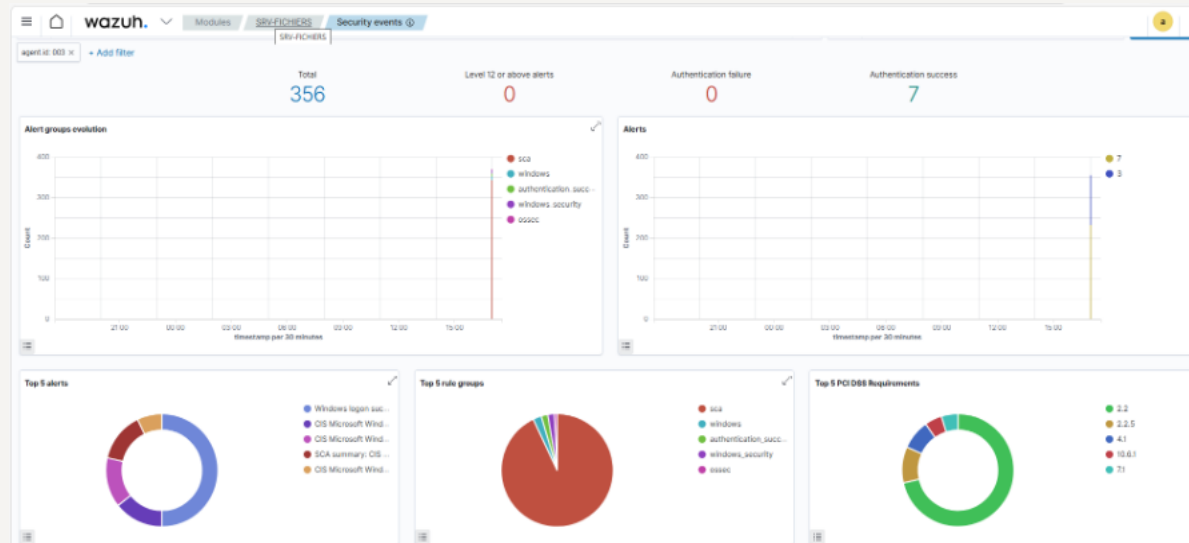
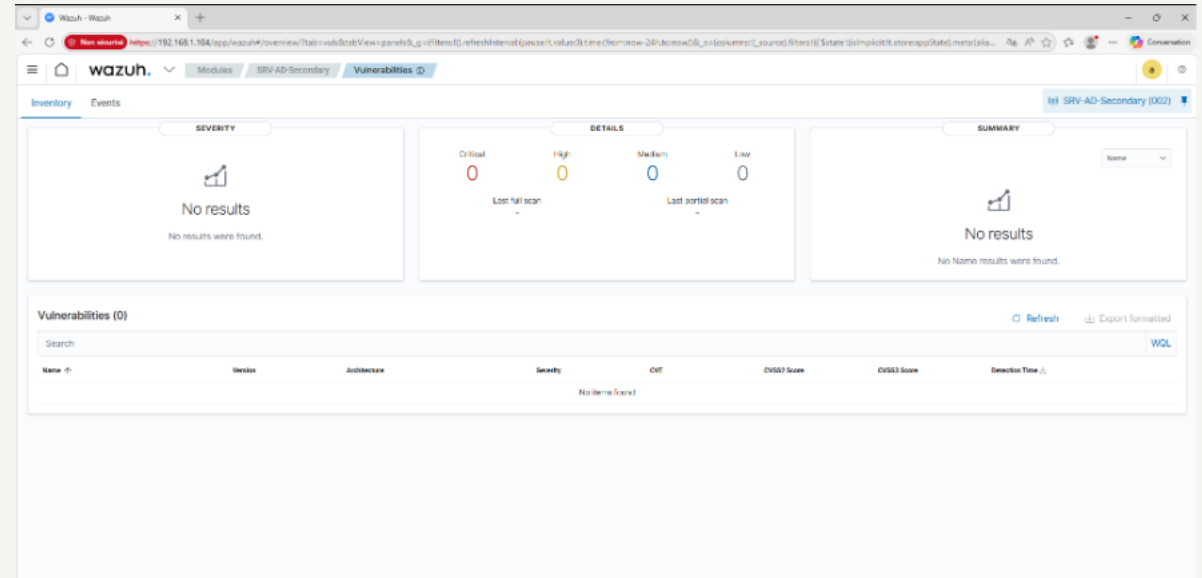
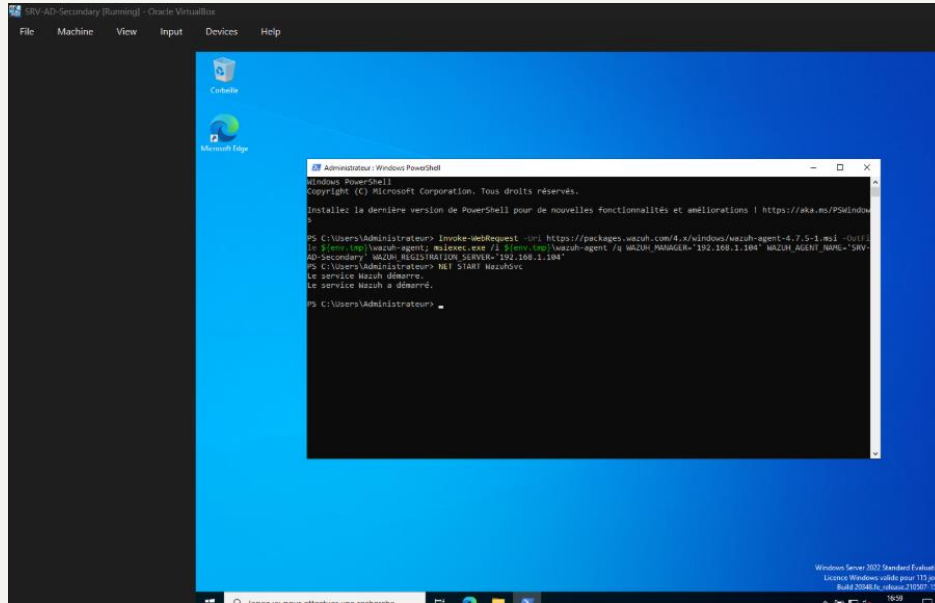
Id : 001

Serveur AD-Primary

système Windows serveur 2022.

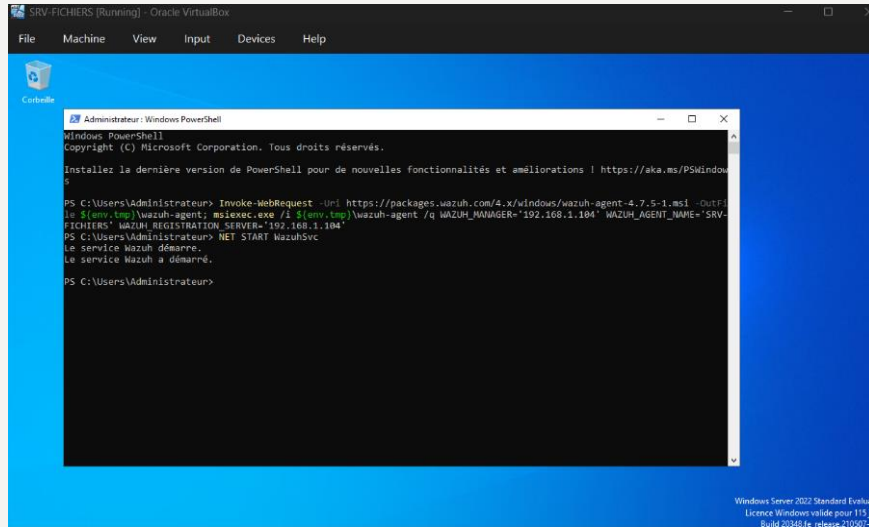
100% agents coverage

Serveur secondary

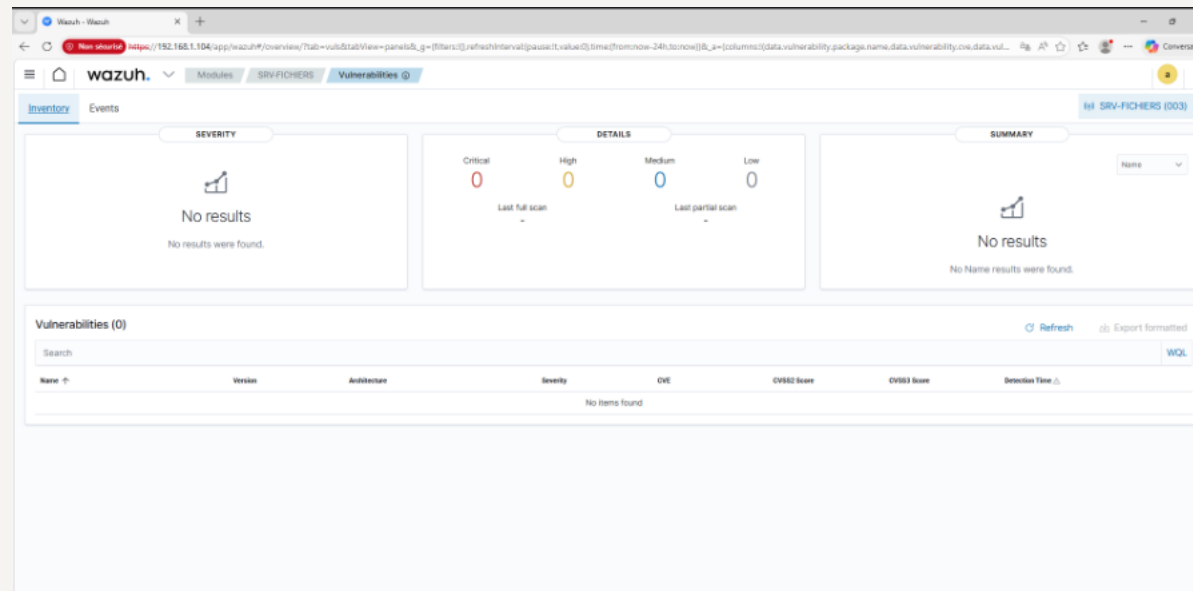
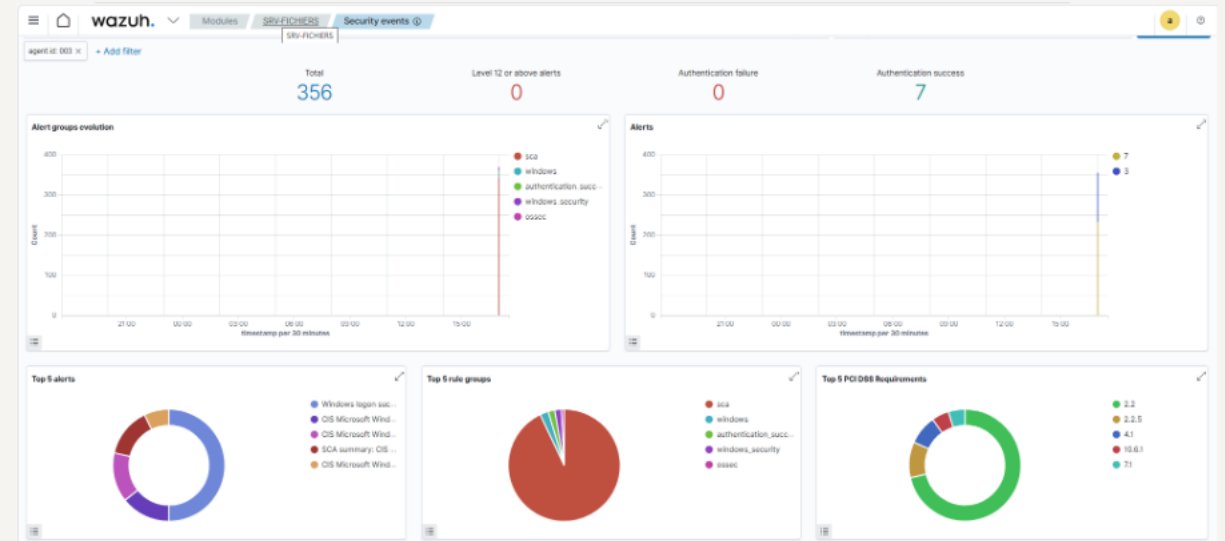


0 vulnérabilités

Serveur fichiers

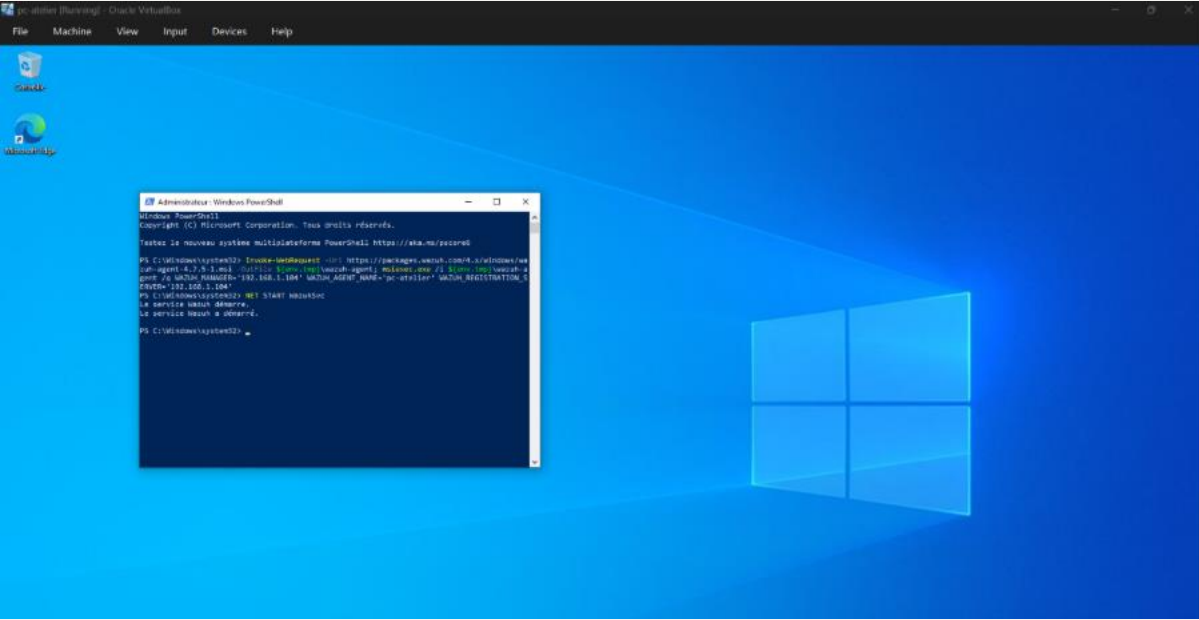


Serveur fichiers

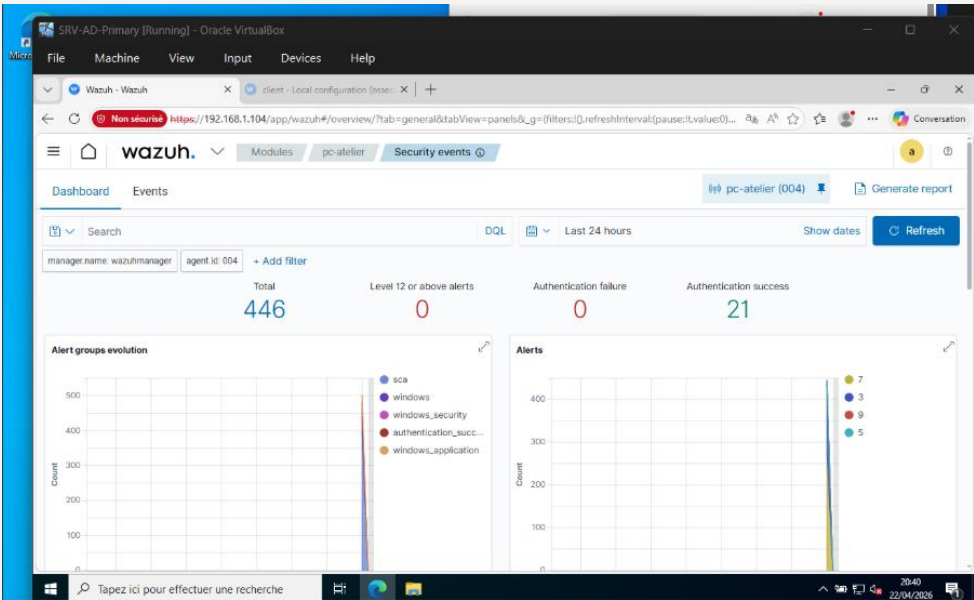
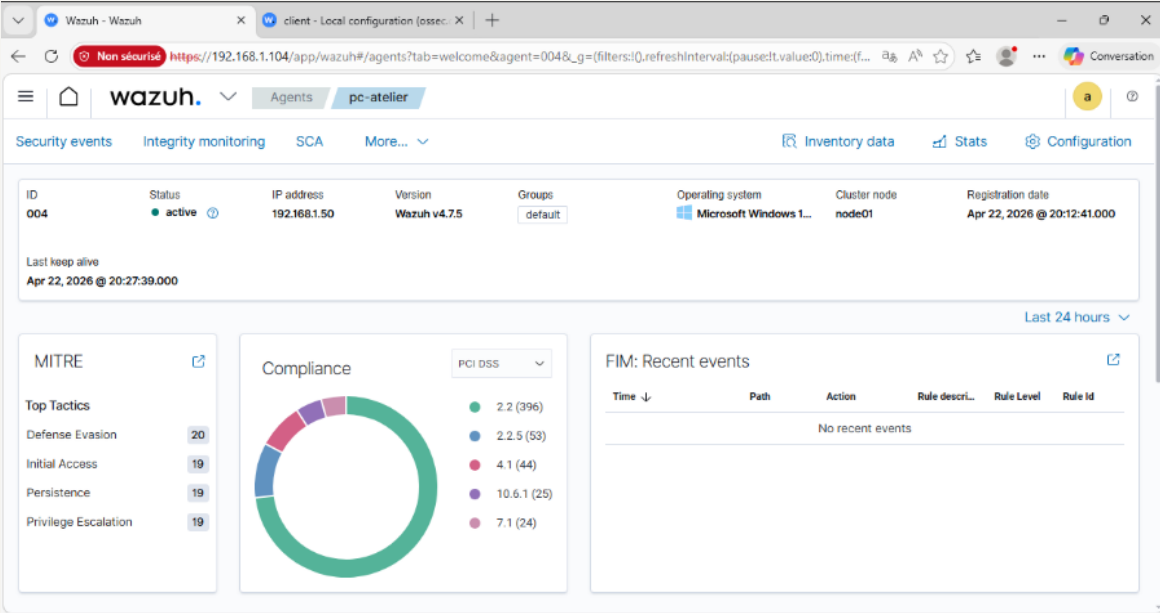


0 vulnérabilités

Pc-Atelier

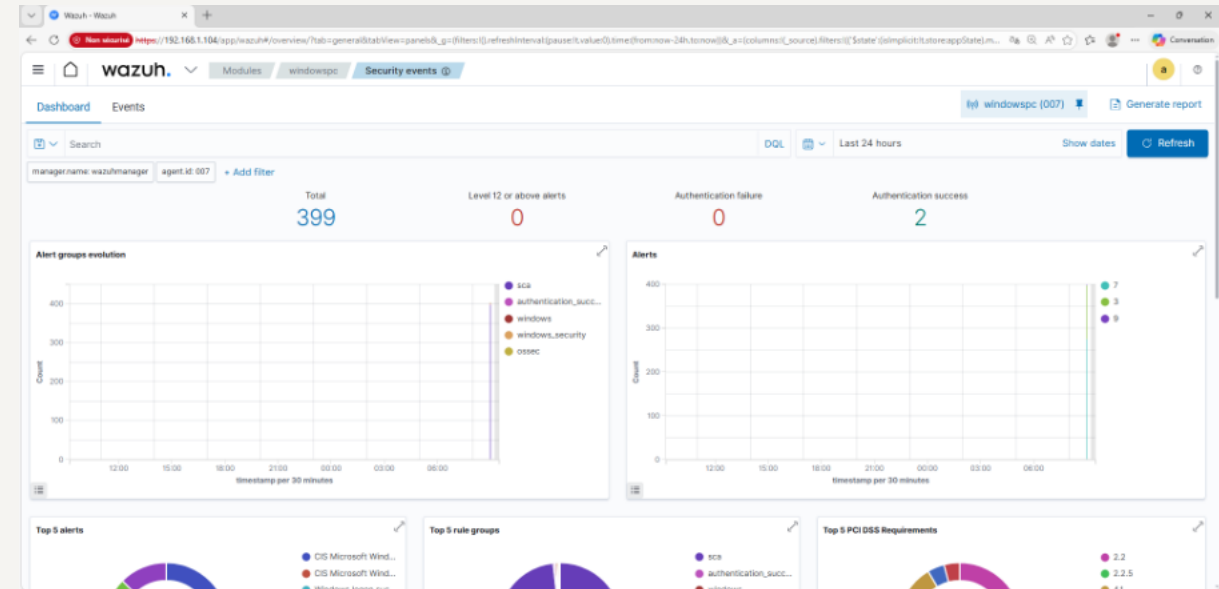
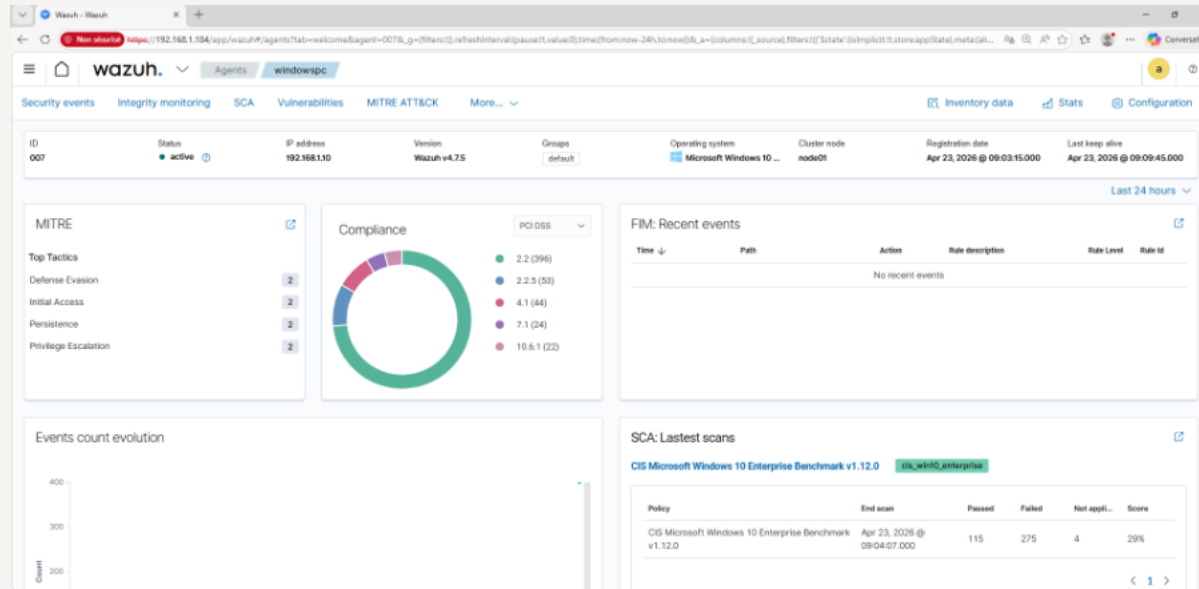


Référentiel-MITRE



0 vulnérabilités

Pc-Admin – 0 vulnérabilité



Test 1 - Détection de Brute Force (Simulation)

SRV-AD-Primary [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Wazuh <https://192.168.1.104/app/endpoints-summary#/agents?tab=welcome&agent=001>

W. [Endpoints](#) **SRV-AD-Primary** API default

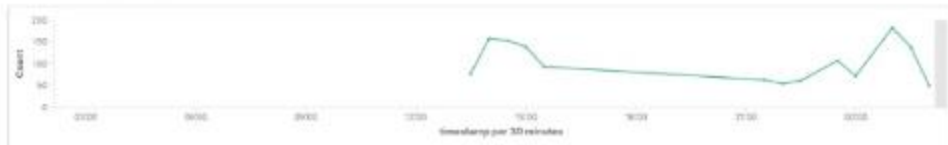
Threat Hunting File Integrity Monitoring Configuration Assessment MITRE ATT&CK Vulnerability Detection More...

by SRV-AD-Primary (001) Stats Conf

ID	Status	IP address	Version	Group	Operating system	Cluster node	Registration date	Last keep alive
001	active	192.168.1.11	Wazuh v4.7.5	default	Microsoft Windows Server 2022 Standard Evaluation X00-20345.587	node01	Apr 22, 2026 @ 15:30:0000	May 16, 2026 @ 02:06:00

(i) Not enough hardware or operating system information

Events count evolution



MITRE ATT&CK

Top Tactics

- Defense Evasion: 004
- Privilege Escalation: 004
- Initial Access: 002
- Persistence: 002
- Execution: 2

Compliance



Vulnerability Detection

0 Critical

0 High

0 Medium

0 Low

Top 5 Packages

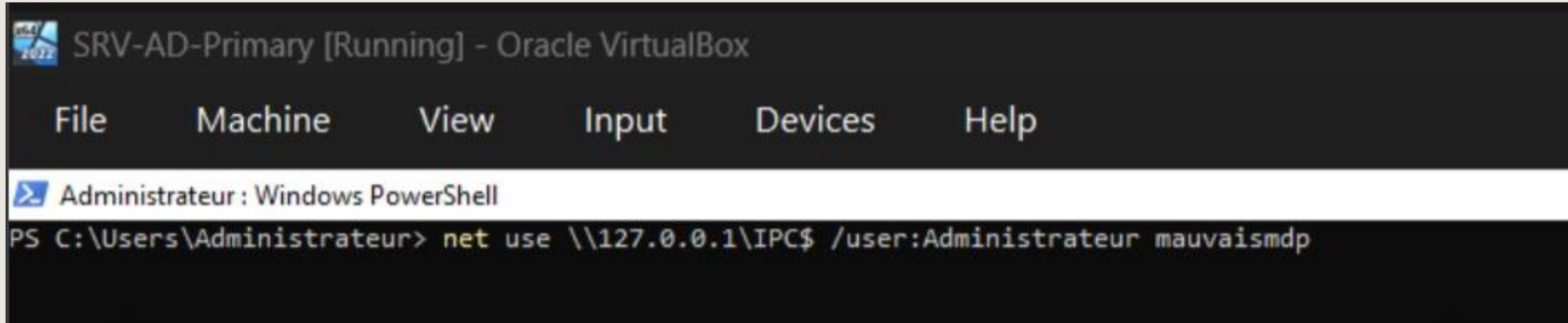
Package: No recent events

Security Configuration Assessment

Policy	End scan	Passed	Failed	Not applicable	Score
OS Microsoft Windows Server 2022 Benchmark v1.0.0	May 16, 2026 @ 01:07:35.000	111	227	4	32%

Échec d'authentification volontaire :

Vérification de la capacité de détection en temps réel sur les tentatives d'accès aux partages administratifs



```
SRV-AD-Primary [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Administrateur : Windows PowerShell
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
```

Nombre de fois que fut exécuté le brute force.

```

Administrateur : Windows PowerShell
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur> net use \\127.0.0.1\IPC$ /user:Administrateur mauvaismdp
L'erreur système 1326 s'est produite.
Le nom d'utilisateur ou le mot de passe est incorrect.
PS C:\Users\Administrateur>

```

Alertes directement émises dans Wazuh

SRV-AD-Primary [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Wazuh

Non sécurisé https://192.168.1.104/app/endpoints-summary#/overview?tab=sca&redirectPolicy=ds_win2022&agentId=0016_g~(filters:[]&refreshInterval(pause:0,value:0).time:(from:now-24h,to:now)&X_a~(filters:[]&query:language:fr...

W. Endpoints SRV-AD-Primary

API default 2

SRV-AD-Primary (001)

Dashboard Inventory Events

CIS MICROSOFT WINDOWS SERVER 2022 BENCHMARK

Passed (111) Failed (227) Not applicable (4)

CIS Microsoft Windows Server 2022 Benchmark v1.0.0

Score 32%

End scan May 16, 2026 @ 01:07:35.000

Checks (342)

Search WQL

ID	Title	Target	Result
27000	Ensure 'Enforce password history' is set to '24 or more passwords'.	Command: net.exe accounts	Not applicable
27001	Ensure 'Maximum password age' is set to '30' or fewer days, but not 0.	Command: net.exe accounts	Failed
27002	Ensure 'Minimum password age' is set to '1' or more day(s).	Command: net.exe accounts	Failed
27003	Ensure 'Minimum password length' is set to '14 or more character(s)'.	Command: net.exe accounts	Failed
27004	Ensure 'Password must meet complexity requirements' is set to 'Enabled'.	Command: powershell Get-ADDefaultDomainPasswordPolicy -Current LoggedOnUser	Passed
27005	Ensure 'Reflex minimum password length limit' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	Failed
27006	Ensure 'Account lockout duration' is set to '15 or more minute(s)'.	Command: net.exe accounts	Failed
27007	Ensure 'Account lockout threshold' is set to '5 or fewer invalid logon attempt(s), but not 0'.	Command: net.exe accounts	Failed
27008	Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: net.exe accounts	Failed
27009	Ensure 'Accounts: Administrator account status' is set to 'Disabled' (MS only).	Command: net user administrator	Failed
27010	Ensure 'Accounts: Block Microsoft accounts' is set to 'Users can't add or log on with Microsoft accounts'.	Registry: HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\System	Failed
27011	Ensure 'Accounts: Guest account status' is set to 'Disabled' (MS only).	Command: net user guest	Failed
27012	Ensure 'Accounts: Limit local account use of blank passwords to console logon only' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Lsa	Passed
27013	Configure 'Accounts: Rename administrator account'.	Command: net user administrator	Failed

Rationale

The Administrator account exists on all computers that run the Windows 2000 or newer operating systems. If you rename this account, it is slightly more difficult for unauthorized persons to guess this privileged user name and password combination. The built-in Administrator account cannot be locked out, regardless of how many times an attacker might use a bad password. This capability makes the Administrator account a popular target for brute force attacks that attempt to guess passwords. The value of this countermeasure is lessened because

Analyse d'une alerte - les dangers du built-in

SRV-AD-Primary [Running] - Oracle VirtualBox

File Machine View Input Devices Help

W. Endpoints SRV-AD-Primary

Checks (342)

ID	Title	Target	Result
27000	Ensure 'Enforce password history' is set to '24 or more password(s)'.	Command: net use accounts	Not applicable
27001	Ensure 'Maximum password age' is set to '365 or fewer days, but not 0'.	Command: net use accounts	Failed
27002	Ensure 'Minimum password age' is set to '1 or more day(s)'.	Command: net use accounts	Failed
27003	Ensure 'Minimum password length' is set to '14 or more character(s)'.	Command: net use accounts	Failed
27004	Ensure 'Passwords must meet complexity requirements' is set to 'Enabled'.	Command: powershell Get-ADDefaultDomainPasswordPolicy -Current LoggedOnUser	Passed
27005	Ensure 'Reset minimum password length limit' is set to 'Enabled'.	Registry: HKLM\LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	Failed
27006	Ensure 'Account lockout duration' is set to '15 or more minute(s)'.	Command: net use accounts	Failed
27007	Ensure 'Account lockout threshold' is set to '5 or fewer invalid login attempt(s), but not 0'.	Command: net use accounts	Failed
27008	Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: net use accounts	Failed
27009	Ensure 'Accounts: Administrator account status' is set to 'Disabled' (MS only).	Command: net user administrator	Failed

Rationale
In some organizations, it can be a daunting management challenge to maintain a regular schedule for periodic password changes for local accounts. Therefore, you may want to disable the built-in Administrator account instead of relying on regular password changes to protect it from attack. Another reason to disable the built-in account is that it cannot be locked out no matter how many failed logons it accrues, which makes it a prime target for brute force attacks. Also, this account has a well-known security identifier (SID) and there are third-party tools that allow authentication by using the SID rather than the account name. This capability means that even if you rename the Administrator account, an attacker could launch a brute force attack by using the SID to log on.

Remediation
To establish the recommended configuration via GP, set the following UI path to Disabled: Computer Configuration\Policies\Windows Settings\Security Settings\Local Policies\Security Options\Accounts: Administrator account status.

Description
This policy setting enables or disables the Administrator account during normal operation. When a computer is booted into safe mode, the Administrator account is always enabled, regardless of how this setting is configured. Note that this setting will have no impact when applied to the Domain Controllers organizational unit via group policy because Domain Controllers have no local account database. It can be configured at the domain level via group policy, similar to account lockout and password policy settings. The recommended state for this setting is: Disabled.

Checks (Condition: any)

- c:\net user administrator /n:Account active=N
- c:\net user administrator /n:The user name could not be found.

Compliance
csc 2.3.1.1
cis, sec: 4.7

ID	Title	Target	Result
27010	Ensure 'Accounts: Block Microsoft accounts' is set to 'Users can't add or log on with Microsoft accounts'.	Registry: HKLM\LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\System	Failed

Running - Oracle VirtualBox

File Machine View Input Devices Help

W. Endpoints SRV-AD-Primary

Checks (342)

ID	Title	Target	Result
27000	Ensure 'Enforce password history' is set to '24 or more password(s)'.	Command: net use accounts	Not applicable
27001	Ensure 'Maximum password age' is set to '365 or fewer days, but not 0'.	Command: net use accounts	Failed
27002	Ensure 'Minimum password age' is set to '1 or more day(s)'.	Command: net use accounts	Failed
27003	Ensure 'Minimum password length' is set to '14 or more character(s)'.	Command: net use accounts	Failed
27004	Ensure 'Passwords must meet complexity requirements' is set to 'Enabled'.	Command: powershell Get-ADDefaultDomainPasswordPolicy -Current LoggedOnUser	Passed
27005	Ensure 'Reset minimum password length limit' is set to 'Enabled'.	Registry: HKLM\LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	Failed
27006	Ensure 'Account lockout duration' is set to '15 or more minute(s)'.	Command: net use accounts	Failed
27007	Ensure 'Account lockout threshold' is set to '5 or fewer invalid login attempt(s), but not 0'.	Command: net use accounts	Failed
27008	Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: net use accounts	Failed
27009	Ensure 'Accounts: Administrator account status' is set to 'Disabled' (MS only).	Command: net user administrator	Failed

Rationale
In some organizations, it can be a daunting management challenge to maintain a regular schedule for periodic password changes for local accounts. Therefore, you may want to disable the built-in Administrator account instead of relying on regular password changes to protect it from attack. Another reason to disable the built-in account is that it cannot be locked out no matter how many failed logons it accrues, which makes it a prime target for brute force attacks. Also, this account has a well-known security identifier (SID) and there are third-party tools that allow authentication by using the SID rather than the account name. This capability means that even if you rename the Administrator account, an attacker could launch a brute force attack by using the SID to log on.

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Checks (Condition: any)

- c:\net user administrator /n:Account active=N
- c:\net user administrator /n:The user name could not be found.

Compliance
csc 2.3.1.1
cis, sec: 4.7

ID	Title	Target	Result
27010	Ensure 'Accounts: Block Microsoft accounts' is set to 'Users can't add or log on with Microsoft accounts'.	Registry: HKLM\LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\System	Failed

Plus de 5 tentatives d'accès effectuées - Blocage du système au bout de la 5eme tentative

Oracle VM VirtualBox

File View Input Devices Help

an sécurisé [https://192.168.1.104/app/endpoints-summary#/overview?tab=sca&redirectPolicy=cis_win2022&agentId=0018_g={filters:\[\]&refreshInterval\(pause:0,value:0\),time:\(from:now-24h,to:now\)}&a={filters:\[\]&query:\(language:fr...](https://192.168.1.104/app/endpoints-summary#/overview?tab=sca&redirectPolicy=cis_win2022&agentId=0018_g={filters:[]&refreshInterval(pause:0,value:0),time:(from:now-24h,to:now)}&a={filters:[]&query:(language:fr...)

Endpoints SRV-AD-Primary

27003	Ensure 'Minimum password length' is set to '14 or more character(s)'.	Command: net.exe accounts	Failed
27004	Ensure 'Password must meet complexity requirements' is set to 'Enabled'.	Command: powershell Get-ADDefaultDomainPasswordPolicy -Current LoggedOnUser	Passed
27005	Ensure 'Reset minimum password length limits' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	Failed
27006	Ensure 'Account lockout duration' is set to '15 or more minute(s)'.	Command: net.exe accounts	Failed
27007	Ensure 'Account lockout threshold' is set to '5 or fewer invalid login attempt(s), but not 0'.	Command: net.exe accounts	Failed
27008	Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: net.exe accounts	Failed

Rationale
Users can accidentally lock themselves out of their accounts if they mistype their password multiple times. To reduce the chance of such accidental lockouts, the Reset account lockout counter after setting determines the number of minutes that must elapse before the counter that tracks failed login attempts and triggers lockouts is reset to 0.

Remediation
To establish the recommended configuration via GP, set the following UI path to 15 or more minute(s): Computer Configuration\Policies\Windows Settings\Security Settings\Account Policies\Account Lockout Policy\Reset account lockout counter after

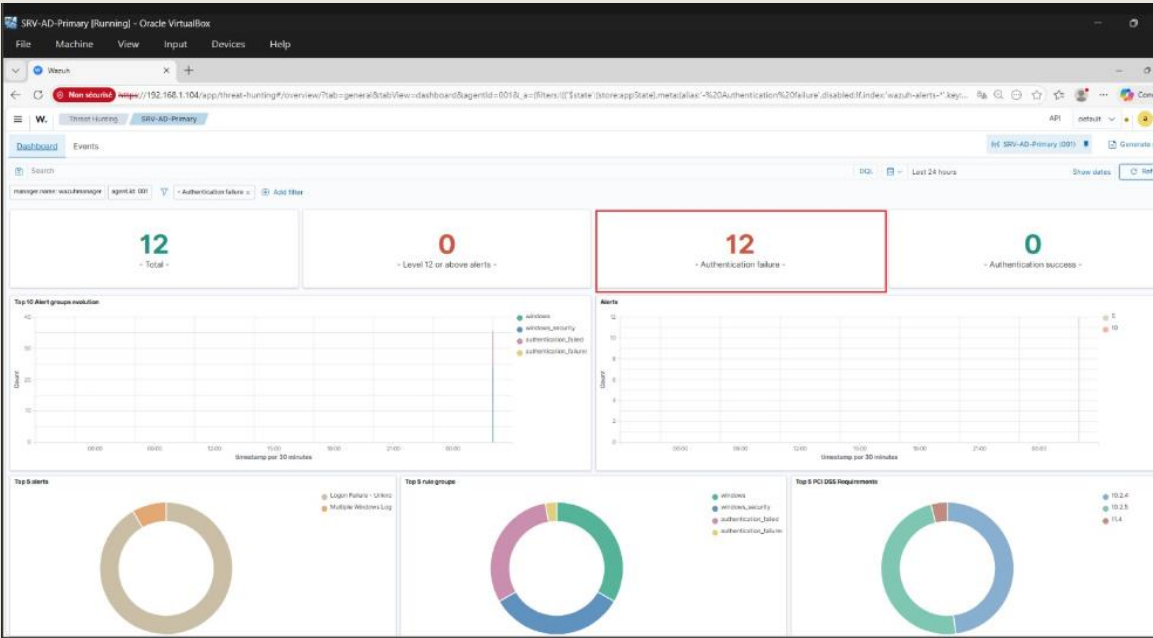
Description
This policy setting determines the length of time before the Account lockout threshold resets to zero. The default value for this policy setting is Not Defined. If the Account lockout threshold is defined, this reset time must be less than or equal to the value for the Account lockout duration setting. If you leave this policy setting at its default value or configure the value to an interval that is too long, your environment could be vulnerable to a DoS attack. An attacker could maliciously perform a number of failed login attempts on all users in the organization, which will lock out their accounts. If no policy were determined to reset the account lockout, it would be a manual task for administrators. Conversely, if a reasonable time value is configured for this policy setting, users would be locked out for a set period until all of the accounts are unlocked automatically. The recommended state for this setting is: 15 or more minute(s). Note: Password Policy settings (section 1.1) and Account Lockout Policy settings (section 1.2) must be applied via the Default Domain Policy GPO in order to be globally in effect on domain user accounts as their default behavior. If these settings are configured in another GPO, they will only affect local user accounts on the computers that receive the GPO. However, custom exceptions to the default password policy and account lockout policy rules for specific domain users and/or groups can be defined using Password Settings Objects (PSOs), which are completely separate from Group Policy and most easily configured using Active Directory Administrative Center.

Check (Condition: all)
• cmd.exe accounts → n:Lockout observation window [(minutes)] is+ (id+) compare >= 15

Compliance
cis: 1.2.3
cis_csc: 4.10

27009	Ensure 'Accounts: Administrator account status' is set to 'Disabled' (MS only).	Command: net user administrator	Failed
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Compteur de verrouillage

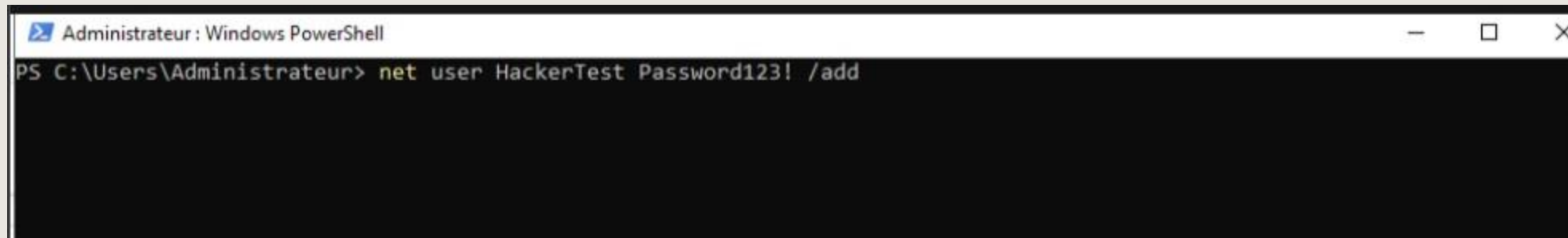


Journalisation des divers évènements

The screenshot displays the Wazuh dashboard with the 'Events' tab selected. The table below shows a list of events with columns for timestamp, agent name, rule description, and rule level.

timestamp	agent name	rule description	rule level	rule
May 16, 2020 @ 02:20:53.7	SRV-AD-Primary	Multiple Windows Logon Failures	10	6002
May 16, 2020 @ 02:20:52.5	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:20:50.1	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:20:47.9	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:20:46.6	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:18:56.5	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:18:38.9	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:17:47.1	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:15:56.7	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:15:54.7	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:15:52.1	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002
May 16, 2020 @ 02:14:29.7	SRV-AD-Primary	Logon Failure - Unknown user or bad password	5	6002

Test 2 - Détection de création d'un compte utilisateur



```
Administrateur : Windows PowerShell
PS C:\Users\Administrateur> net user HackerTest Password123! /add
```

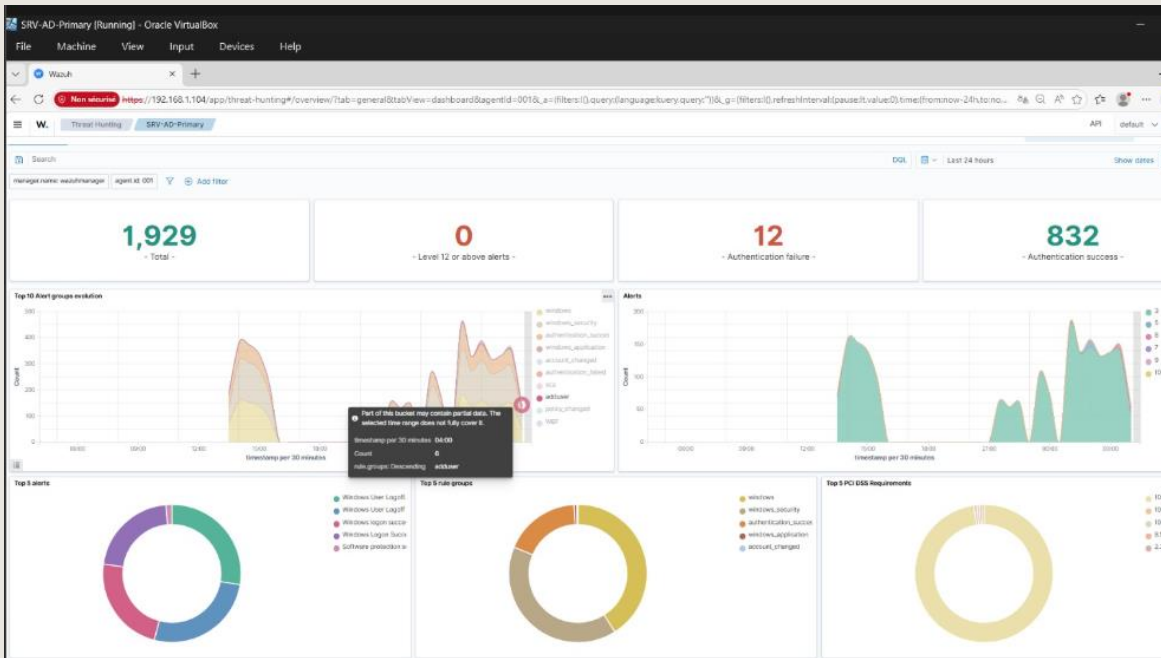


```
Administrateur : Windows PowerShell  
PS C:\Users\Administrateur> net user HackerTest Password123! /add  
La commande s'est terminée correctement.  
PS C:\Users\Administrateur> _
```

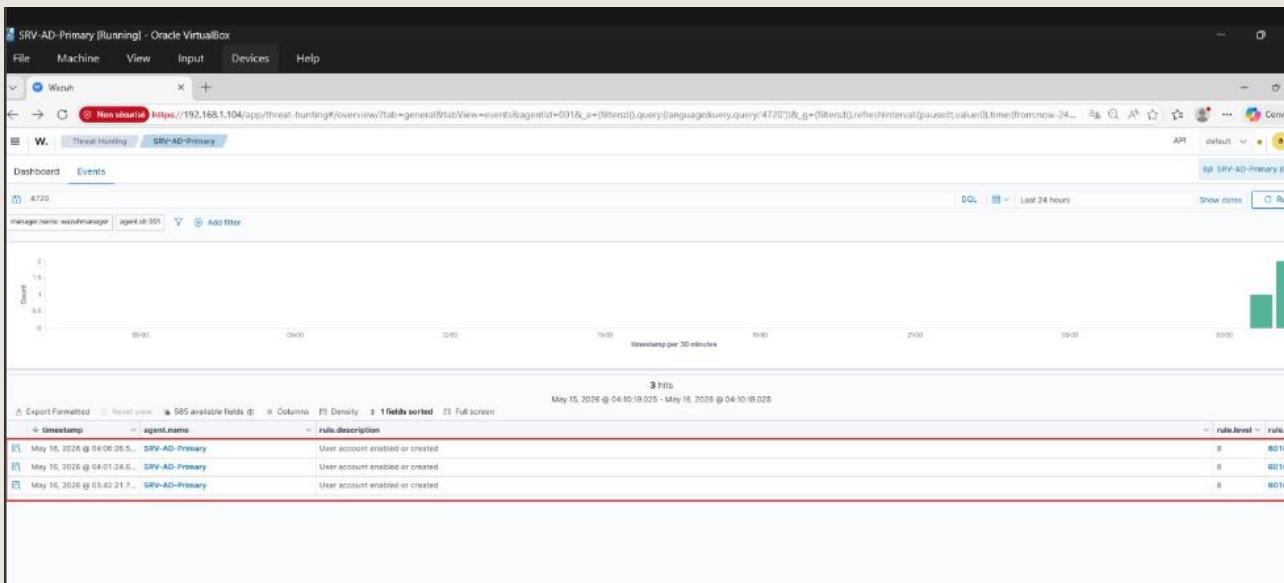
Aller dans Threat Hunting ou Security Events
Filtrer par agent : SRV-AD-Primary

Event - barre de recherche : 4720.

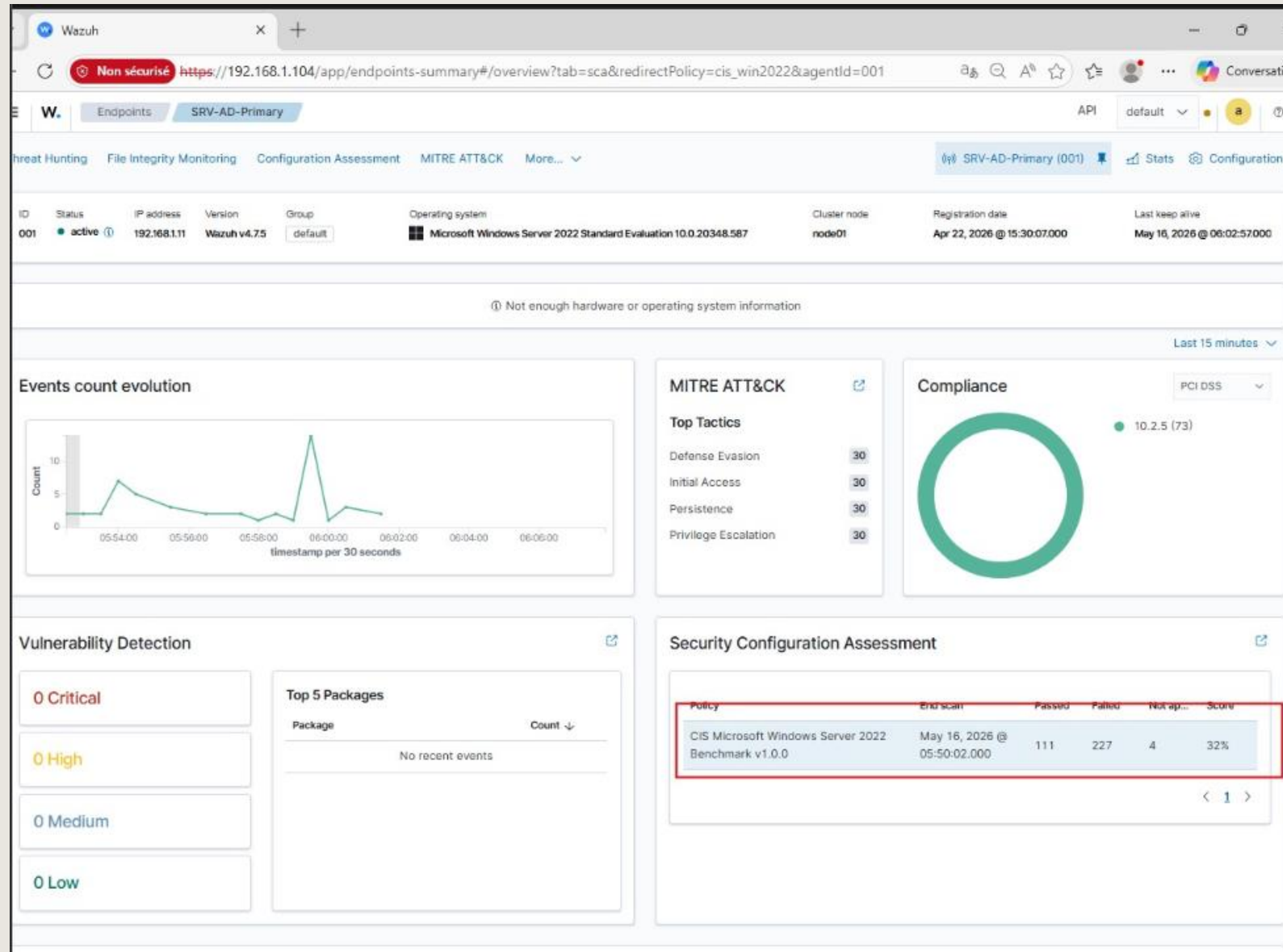
Identifiant d'évènement de la création de compte Windows.



Détection de création de compte réussi.



Test 3 - Analyse de conformité SCA (CIS Benchmark)



Politique analysée :
CIS Microsoft
Windows Server
2022 Benchmark
v1.0.0

Vérifications
réussies : 111

Vérifications
échouées : 227

Score de
conformité : 32%

